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Evolutionary remodeling of non-canonical ORF translation in mammals

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eLife Assessment

This study presents a large, systematically curated catalog of non-canonical open reading frames (ncORFs) in human and mouse by reanalyzing nearly 400 Ribo-seq datasets using a standardized pipeline; the resulting atlas consolidates ncORF annotations across tissues and provides a **valuable** reference for understanding non-canonical translation and ORF emergence. The main conclusions are supported by consistent data processing and multiple computational measures of translation and conservation. While the pipeline is transparent and robust, several downstream analyses are descriptive, and some evolutionary interpretations remain correlative; dataset heterogeneity, uneven tissue representation, and limited experimental validation also constrain the strength of a subset of the findings. Overall, the evidence is **solid**, and the resource will be broadly used by the community.

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Abstract

Non-canonical open reading frames (ncORFs) are pervasive within transcripts annotated as “non-coding” or “untranslated regions” of mRNAs, yet their landscape under normal physiological conditions remains to be fully resolved, particularly outside humans. Here we applied a stringent and standardized pipeline to hundreds of high-quality ribosome profiling libraries from normal mammalian tissues and cell types, identifying 11,623 human and 16,485 mouse ncORFs. Evolutionary analyses revealed that thousands of ncORFs are subject to coding constraint and exhibit lineage-specific conservation, underscoring their functional potential. Ancient ncORFs are preferentially highly translated, broadly expressed, and enriched for lineage-specific conservation. Co-expression patterns further indicate that many ncORFs, especially ancient ones, are cotranslated with canonical coding sequences, consistent with functions mediated through protein–protein interactions. Together, these findings establish a comprehensive atlas of mammalian ncORFs and provide fundamental insights into their evolutionary dynamics and functional integration within the proteome.

Introduction

The systematic identification of functional genomic elements represents a foundational objective in the field of genomics. Among these elements, non-canonical open reading frames (ncORFs)—which are open reading frames (ORFs) located within long non-coding RNAs (lncRNAs) or untranslated regions (UTRs) of mRNAs—have attracted considerable attention in recent years^{1–5}. While certain ncORF subtypes, such as upstream ORFs (uORFs) located in 5' UTRs and downstream ORFs (dORFs) in 3' UTRs, have been extensively characterized for their ability to regulate mRNA translation and stability *in cis*, a growing body of evidence demonstrates that ncORFs can also give rise to functional proteins that engage in diverse cellular pathways^{1,2,6,7}. As functional ncORFs continue to be discovered across multiple eukaryotes^{8–17}, systematic elucidation of their composition and function promises to uncover an additional layer of regulatory complexity and to refine our conception of the so-called “non-coding” genome.

Although ncORFs may play important biological roles, the number of translated ncORFs in the human genome remains uncertain. Detection by mass spectrometry (MS)-based proteomics is challenging, likely due to their short lengths and low abundances¹⁸. Ribosome profiling (Ribo-Seq) offers greater sensitivity for identifying translation events^{18,19}, but downstream computational pipelines often produce highly divergent ncORF sets²⁰. Consequently, published estimates of ncORF abundance span several orders of magnitude—ranging from thousands to millions^{15–22}. To establish a standardized reference, the GENCODE consortium integrated data from seven distinct studies to produce a catalog of 7,264 human ncORFs (hereafter GENCODE ncORFs)²; however, the heterogeneity of sample sources and prediction algorithms likely introduced systematic bias. This catalog is also restricted to AUG-initiating ORFs, thereby overlooking those with near-cognate start codons. A later effort generated a high-resolution human translome map and identified 7,767 ncORFs with a unified pipeline²². Nevertheless, this conservative annotation disproportionately under-represents coding sequence (CDS)-overlapping ncORFs. Indeed, later estimation suggests that between 10,500 to 22,500 ncORFs should exist in the human genome¹⁸. Furthermore, many ncORFs were reported from studies of cancer cells^{1,13,23,24}, where extensive transcriptome reprogramming alters the translational landscape^{25,26}. While paired healthy tissues were also analyzed in these studies, the landscape of ncORFs translated under normal physiological conditions is not yet fully resolved.

While a limited subset of ncORFs has been shown to encode signaling molecules, cofactors, membrane proteins, or subunits of large protein complexes^{1,27,28}, functions of most ncORF-encoded proteins (ncEPs) are largely uncharacterized. Advances in high-throughput screen technologies have enabled the systematic interrogation of functional ncEPs^{23,28,30–33}. Nevertheless, there is minimal concordance among hits from different studies³. Evolutionary analyses can offer insights into potential functionality of these translation products and mechanisms of origination^{34–36}, yet how ncORF translation has been remodeled to integrate into proteome since their evolutionary origins is still poorly understood. Consequently, the extent to which translated ncORFs encode functional proteins and the evolutionary dynamics of their integration into the existing proteome are largely unresolved. Moreover, because most ncORF studies have focused on humans, it is unclear whether ncORFs in other species share similar sequence features and expression patterns.

To address these gaps, we generated a standardized annotation of translated ncORFs by uniformly reprocessing ~400 high-quality ribosome profiling libraries from normal mammalian tissues and cell types. This stringent approach yielded 11,623 human and 16,485 mouse ncORFs, consistently recovered across diverse samples and prediction strategies. Comparative and functional genomic analyses revealed that thousands of these ncORFs are under evolutionary constraint and exhibit distinct translation profiles. The resulting dataset provides a robust, comprehensive resource that will facilitate future investigations into the biological roles of ncORFs across cellular and organismal contexts.

Results

Annotation of ncORFs translated in normal mammalian cells

We developed a standardized pipeline to systematically annotate ncORFs translated in normal human and mouse cells (Fig. 1A [↗](#)). We first curated publicly available Ribo-Seq libraries, excluding those from pathological or genetically modified samples. Since both library size and the proportion of in-frame ribosome-protected fragments (RPFs) affect the sensitivity and accuracy of ncORF prediction²⁰, we filtered out libraries containing fewer than 5 million RPFs or with in-frame rates below 60%. After this quality control step, we retained 174 human and 209 mouse high-quality Ribo-Seq libraries, spanning a range of tissues and cell types (Fig. S1 [↗](#); Table S1 [↗](#)). We predicted translated ORFs that begin with NUG (where N stands for any nucleotide) start codons and encode at least five amino acids using three tools—PRICE³⁷, RiboCode³⁸ and Ribo-

TISH³⁹—which we previously benchmarked for optimal performance²⁰. After excluding annotated CDSs and their extension or truncation variants arising from alternative initiation sites, we obtained a preliminary list of ncORFs in each library.

To generate reliable ncORF annotations, we applied stringent filtering and integration criteria. We removed potential false positives caused by non-ribosomal contamination or translational noise based on metrics including Fragment Length Organization Similarity Score (FLOSS)⁴⁰, Ribosomal Release Score (RRS)^{41,42} and the number of RPFs within an ORF by P-site. We then aggregated the remaining ORFs to identify those consistently detected across multiple methods and libraries in human and mouse datasets, respectively (Fig. 1A). In many cases, two ORFs predicted from different libraries or methods may share the same start or stop codon due to the alternative translation initiation or splicing, resulting in groups of overlapping ncORFs. For example, 11.1% (805) of GENCODE ncORFs overlap with at least one other ORF, forming 329 overlapping groups (Table S2). This overlap complicates the identification of reproducible ORFs, as ORFs within a group may represent distinct proteoforms of a protein. To address this issue, we collapsed each group of ncORFs sharing the same start or stop codon into a single cluster, and retained only those clusters supported by at least two independent methods and observed in more than one library. For each retained cluster, we selected a representative ORF by prioritizing transcript annotation^{43,44}, start codon type, and ORF length to facilitate downstream analyses (Fig. 1A).

This approach yielded 12,469 translated ncORFs in humans and 16,960 in mice. However, manual inspection revealed a subset of spurious ncORFs that are not truncation or extension variants of annotated CDSs, but instead partially overlap with annotated CDSs from alternative transcript isoforms in frame (Figs. 1B & C). For instance, in gene *AGTPBP1*, the CDS of transcript *ENST00000357081* begins downstream of *ENST00000628899* CDS due to the inclusion of an alternative exon. As a result, the 5' UTR of *ENST00000357081* appears to contain a translated uORF (Fig. 1B), which overlaps in-frame with the CDS of *ENST00000628899* and may be mistakenly identified as a translated ncORF owing to active translation of the latter. A similar case is seen with the lncRNA ORF (lncORF) in *ENST00000462211*, which overlaps in-frame with the CDS of *ENST00000328767* from the *TMTC4* gene and may thus be erroneously identified as a translated ncORF (Fig. 1C). To avoid such artifacts, we implemented an additional filtering step to exclude these overlapping cases and left with 11,623 and 16,485 high-confidence ncORFs in humans and mice, respectively (Table S3).

Annotated ncORFs are supported by independent evidence

To assess whether the annotated ncORFs are supported by independent evidence, we compared them with a previously compiled catalog of ncORFs identified using mass spectrometry (MS)-based proteomic data²⁰. For the 3,494 MS-supported ncORFs in humans and 873 in mice, 44.5% and 67.6%, respectively, were also predicted as translated ncORFs in this study (Fig. S2). We further evaluated our annotation against the GENCODE ncORFs², a widely used community resource. Of the 7,264 GENCODE ncORFs, 6,974 are located in fully annotated transcripts and were included in our analysis. Among them, 2,997 were reported by at least two independent studies (hereafter “phase 1”), while the remaining were only reported in a single study (“single”)². We found that 75.6% of phase 1 ncORFs were independently recovered in our annotation, significantly more than the single-study ncORFs (Fig. 1D; $P = 6.4 \times 10^{-307}$, Fisher’s exact test).

A recent study conducted a comprehensive pan-proteome analysis to provide high-confidence peptide evidence for GENCODE ncORFs⁴⁵. In this study, ncORFs were categorized into four tiers: tier 1 and 2 ncORFs had support from both MS and Ribo-Seq data; tier 3 had MS evidence only; and tier 4 had Ribo-Seq evidence only²⁹. Notably, our annotation was significantly enriched for tier 1 or 2 ncORFs ($P = 2.0 \times 10^{-64}$, Fisher’s exact test), with 446 and 661 of them detected, respectively (Fig. 1E). Combining MS-supported ncORFs from our previous compilation and those from GENCODE ncORFs, 2,088 (18.0%) human ncORFs identified in this study were supported by MS evidence. Despite this overlap, the majority (65.2%) of human ncORFs in our

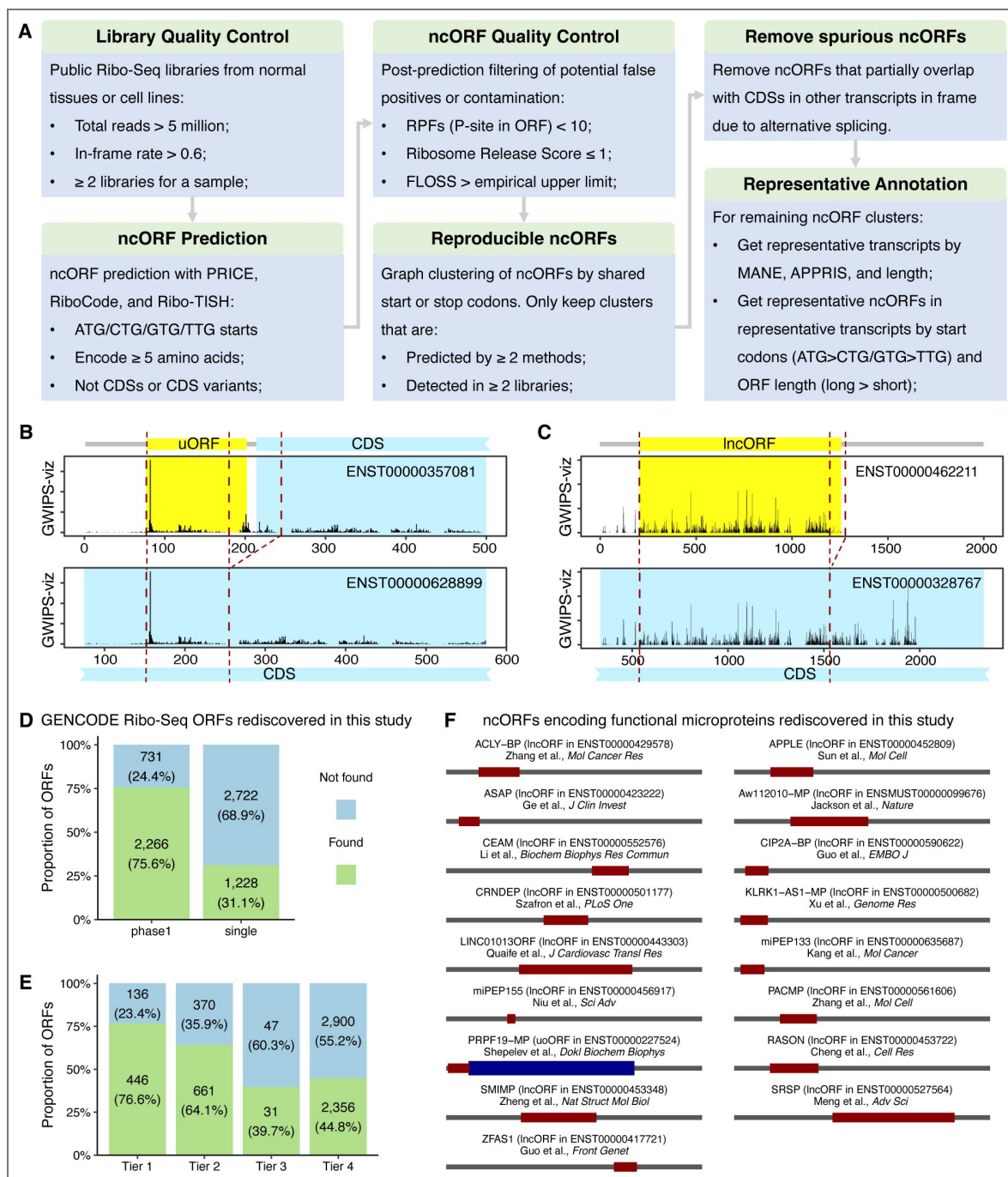


Figure 1. Annotation of ncORFs in humans and mice.

(A) Overview of the integrative pipeline for ncORF annotation in human and mouse genomes. (B-C) RPF coverage plot for a representative uORF (B) and a lncORF (C), visualized using the Ribo-Seq signal track from GWIPS-viz⁶⁸. ncORFs were shown in yellow and CDSs in cyan. Dashed lines indicate identical genomic regions shared across transcript isoforms. (D-E) Proportion of GENCODE ncORFs rediscovered in this study. GENCODE ncORFs were stratified either by reproducibility across independent studies (D) or by tiered translation evidence strength (E). (F) Known ncORFs that were experimentally characterized in earlier studies and independently rediscovered in this study from humans and mice. CDS, coding sequence; FLOSS, fragment length organization similarity score; MANE, matched annotation from NCBI and EBI; ncORF, non-canonical open reading frame; ORF, open reading frame; RPF, ribosome-protected fragment.

annotation are not included in either of the two catalogs. This highlights both the complexity of the non-canonical translome and the critical role of tissue diversity in comprehensive ncORF annotation.

To further validate our findings, we manually curated ncORFs with known functions from the literature. We identified 17 ncORFs in our annotation that have been experimentally confirmed to be translated—either through reporter assays or antibody detection—and shown to produce functional proteins (Fig. 1F [↗](#); Table S4 [↗](#)). Notably, 16 of these validated ncORFs are from humans, and only 12 are included in the GENCODE catalog. This analysis further underscores the accuracy and functional relevance of our annotations. Together, these results indicate that our annotation pipeline generates a robust and reliable catalog of translated ncORFs in mammals.

ncORFs exhibit signatures of translation and encode putative proteins lacking known domains

We classified ORFs into categories based on community recommendations² and found that the distribution of ORF types closely aligns with previous estimates¹⁸. In both human and mouse, uORFs, upstream overlapping ORFs (uoORFs), and lncORFs are more prevalent than other types, whereas internal ORFs (iORFs), downstream overlapping ORFs (doORFs), and dORFs are relatively rare (Fig. 2A [↗](#)). Although most ncORFs are short, encoding fewer than 100 amino acids (Fig. 2B [↗](#)), their sequences show clear signatures of translation similar to CDSs. These include preferential usage of AUG start codons (Fig. S3A [↗](#)), enrichment toward 5' transcript ends (Fig. S3B [↗](#)), and reduced RNA secondary structure⁴⁶ (Fig. S3C [↗](#)) and optimized Kozak contexts^{47,48} around start codons, especially in non-AUG ncORFs (Figs. S3D [↗](#) & S4 [↗](#)). Despite these similarities, ncORFs differ markedly from CDSs in both codon and amino acid usage (Figs. S5A–B [↗](#)). This divergence is further supported by their lower tRNA adaptation index values and higher effective number of codons, indicating less optimized codon usage (Figs. S5C–D [↗](#)). Together, these findings show that ncORFs share core sequence and structural features with canonical CDSs, yet maintain distinct codon and amino acid usage profiles.

Although ncORFs can encode putative proteins, the functional significance of ncEPs remains poorly understood. To assess their potential functions, we searched for known protein domains within their sequences. Overall, only 1.29% (150) of human and 0.78% (129) of mouse ncEPs contained at least one known domain (Fig. 2C [↗](#); Table S5 [↗](#)), presumably due to their short lengths relative to canonical proteins. Consistently, domain-containing ncORFs are significantly longer than those without annotated domains (Fig. S6 [↗](#)) and are predominantly classified as lncORFs (Fig. 2D [↗](#)). Unlike canonical proteins, which have relatively low intrinsically disordered region (IDR) content (median $\pm$ standard error: $14.6 \pm 0.28\%$ in humans and $12.7 \pm 0.26\%$ in mice), ncEPs exhibit significantly higher IDR proportions (Fig. 2E [↗](#)), with medians of $62.5 \pm 1.42\%$ and $60.0 \pm 1.27\%$, respectively. These results indicate that most ncEPs lack structured domains and are instead dominated by IDRs.

Among the eight domains observed in at least five ncORFs, three are DNA-binding domains: zinc finger, Krüppel-associated box, and helix-turn-helix (Fig. 2F [↗](#)). We hypothesize that these domains might originate from transposable elements (TEs), which often encode DNA-binding proteins involved in transposition⁴⁹. Indeed, domain-containing ncORFs were significantly enriched for TE overlap in both humans and mice ($P = 5.8 \times 10^{-4}$ and 3.4×10^{-17} , respectively; Fisher's exact tests; Fig. 2G [↗](#)). Together, these results suggest that only a subset of longer ncORFs have acquired modular domains, potentially through TE-related mechanisms.

Putative ncEPs are selectively constrained

Given the scarcity of known domains in putative ncEPs, a key question is whether they are functional in cells. If at least some of these ncEPs are not merely translational noise or byproducts of translation-dependent functions, we would expect them to exhibit evolutionary signatures of selective constraints similar to canonical CDSs. To test this, we leveraged the recent high-resolution genomic constraint maps derived from both intra- and inter-species comparisons in

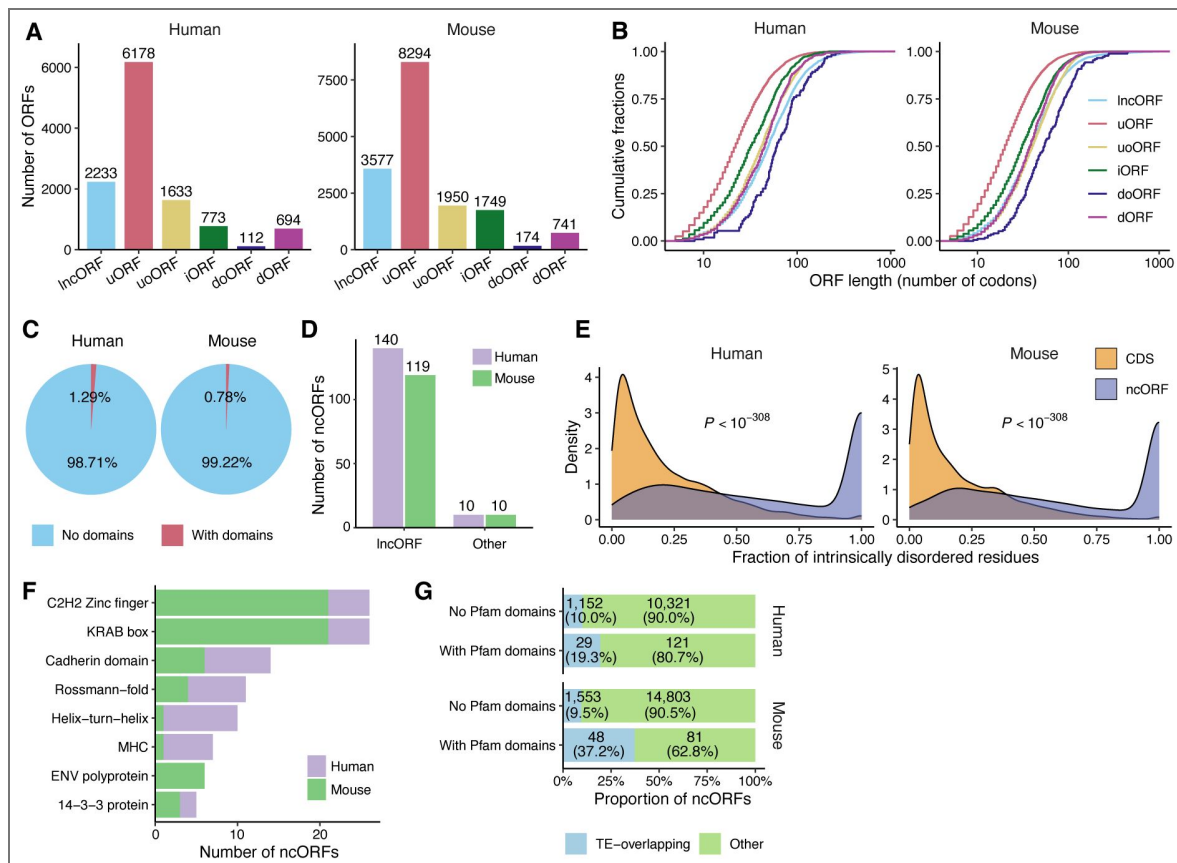


Figure 2. Sequence features of ncORFs and putative ncEPs in humans and mice.

(A) Number of ncORFs belonging to different categories. (B) Distribution of ncORF lengths. (C) Proportion of ncEPs that contain known Pfam domains. (D) Number of ncORFs with Pfam domains among IncORFs and other ncORF categories. (E) Distribution of the proportion of intrinsically disordered residues in CDS- and ncORF-encoded proteins. (F) Most frequently identified Pfam domains among human and mouse ncORFs. (G) Proportion of ncORFs overlapping with TEs, stratified by presence or absence of Pfam domains. Differences were assessed using Wilcoxon rank-sum tests. CDS, coding sequence; ncORF, non-canonical open reading frame; ncEP, ncORF-encoded protein; TE, transposable element.

humans^{50,51}. First, we examined the genomic non-coding constraint of haploinsufficient variation (Gnocchi), which quantifies selective pressure by comparing observed versus expected variants across human genomes. A Gnocchi score above zero indicates purifying selection⁵⁰. Although Gnocchi scores of ncORFs are lower than those of CDSs, they are significantly greater than zero (Fig. S7⁴), suggesting that ncORF sequences are under evolutionary constraint. A hallmark of functional coding regions is codon position-specific constraints, but Gnocchi scores lack the resolution needed to assess such fine-scale features. Therefore, we next analyzed the base-resolution conservation metric PhyloP⁵² calculated from multiple alignments of primate or mammalian genomes⁵¹. As expected, PhyloP scores in CDSs are substantially higher than those in flanking UTRs and display strong three-nucleotide periodicity, with the third codon position being less conserved than first and second (Fig. 3A⁴). Although ncORFs (excluding those overlapping annotated CDSs) do not show higher overall PhyloP than flanking regions, they also exhibit significant three-nucleotide periodicity in both mammals (Fig. 3A⁴) and primates (Fig. S8⁴), suggesting that at least a subset of ncORFs are subject to similar evolutionary constraints as CDSs.

Motivated by the above results, we next sought to identify individual ncORFs under selective pressure. To this end, we determined their modes of origination^{34,35} and evolutionary history through comparative genomic analyses. Our analysis revealed that most ncORFs are evolutionarily young, with 63.0% (7,319) and 82.8% (13,508) ncORFs in humans and mice originating within the mammalian lineage, respectively (Figs. 3B⁴ & S9⁴). In line with previous studies^{34,35}, we estimate that at least 51.7% of human and 50.8% of mouse ncORFs arise *de novo* (Table S6⁴). We also observed that the number of ncORFs originating at each ancestral node, when normalized by branch length, decreases with increasing node-to-tip distance (Fig. 3C⁴), reflecting dynamic turnover and reduced long-term retention of ncORFs. To assess whether individual ncORFs are constrained for coding, we analyzed codon substitution patterns within ncORFs since origination³⁴ with PhyloCSF⁵³. PhyloCSF estimates the log-likelihood ratio of a given alignment under coding versus non-coding models; positive values indicate stronger similarity to conserved coding sequences. We found 13.8% (1,608) of human and 32.0% (5,281) of mouse ncORFs have positive PhyloCSF scores (Fig. 3D⁴), and 10.9% (1,264) of human and 22.6% (3,721) of mouse ncORFs even have PhyloCSF exceeding 10 (Table S6⁴), providing strong evidence that these sequences may encode functional ncEPs.

Evolutionary dynamics can also provide insights into the function of ncORFs, as functionally constrained ncORFs are less likely to be lost over time. To quantify the conservation of each ncORF since its origination, we developed a local branch length score (BLS) (Fig. 4A⁴), adapting the “global” BLS metric previously used to assess uORF conservation across full phylogenies^{48,54}. Unlike global BLS, local BLS is designed to detect lineage-specific constraint, making it better suited for identifying functional ncORFs that may be restricted to certain evolutionary branches. Using a local BLS threshold of 0.9, we identified 16.3% (1,889) of human and 24.8% (4,087) of mouse ncORFs as exhibiting lineage-specific conservation (Figs. 4B⁴ & S9⁴). Notably, these ncORFs are significantly more likely to show positive PhyloCSF scores (Fig. 4C⁴), reflecting the effectiveness of local BLS. In total, we identified 681 human and 1,622 mouse ncORFs that are both lineage-specifically conserved and show evidence of coding potential, making them strong candidates for future functional validation. Together, these findings suggest that a subset of ncORFs may encode functional proteins rather than translational byproducts.

Expression of ncORFs is shaped by their evolutionary dynamics

Gene expression is a tightly regulated process, and non-functional products are unlikely to be robustly expressed. To assess the expression of ncORFs, we examined their translation across diverse tissues and cell lines (Table S7⁴). On average, 71.9% of human and 48.5% of mouse ncORFs are translated with reads per kilobase per million mapped reads (RPKM) ≥ 1 , which is comparable to or slightly exceeds the proportions observed for CDSs (Fig. S10⁴). To investigate the influence of ncORF evolution on their expression, we analyzed mean translation levels of ncORFs across different samples. We found that ncORFs with higher local BLS tend to be more highly translated in both species (Fig. S11A⁴), whereas coding potential showed only a minor

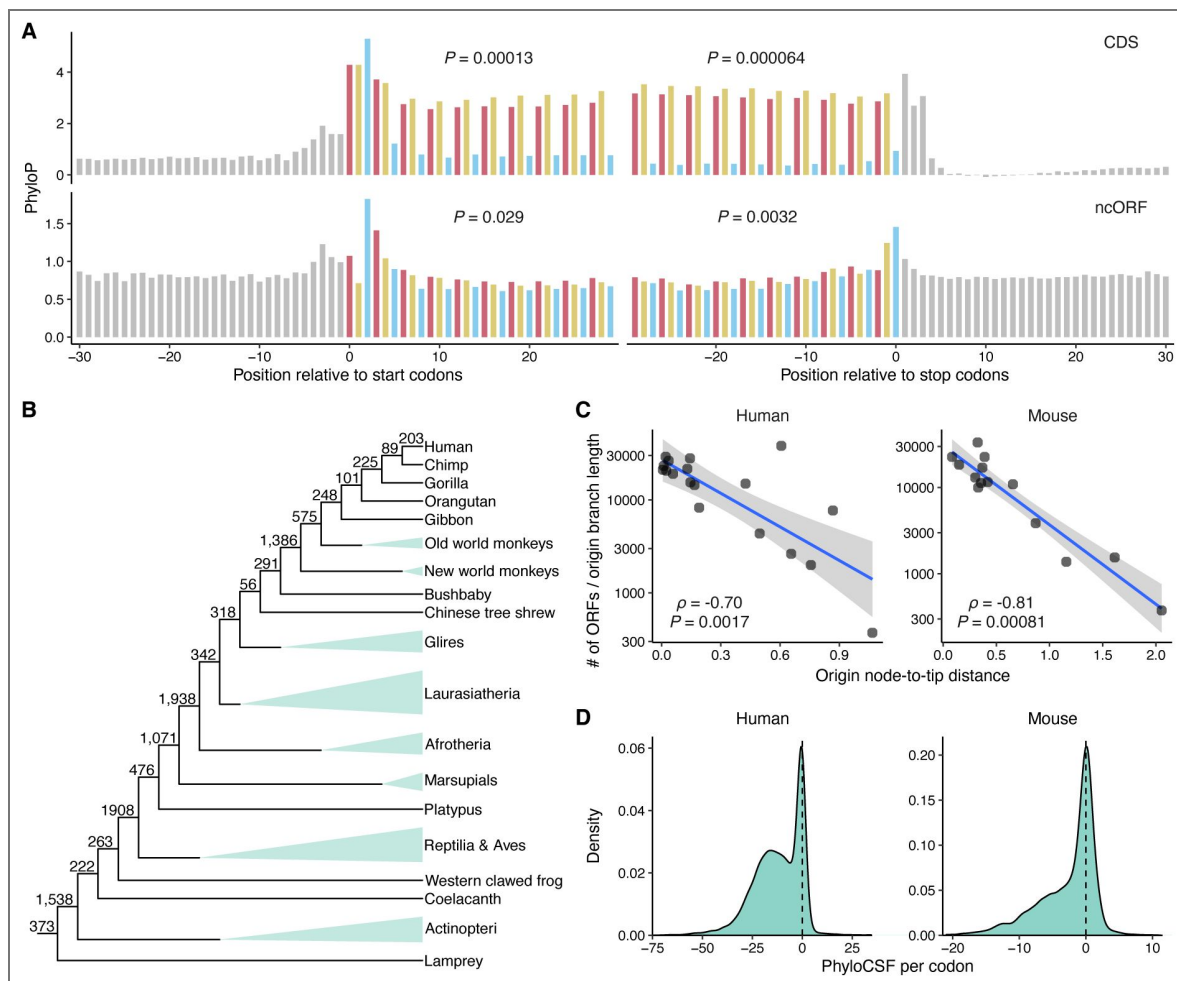


Figure 3. Evolutionary constraints of ncORFs.

(A) The mean PhyloP scores of 30 base pairs upstream and downstream of the ORF start or stop codons in mammals. Codons were delineated with bars of alternating colors by different frames, and nucleotides in untranslated regions were shown in grey. The significance of three-nucleotide periodicity was assessed by autocorrelation with a lag of three. (B) Cladograms of vertebrates with the number of ncORFs originating at each ancestral branch of humans. Triangles indicate species merged into larger clade for visual simplicity. (C) Relationship between ORF origination rates and node ages measured as node-to-tip distances. Origination rate was defined as the number of ncORFs that originated on a branch divided by the branch length. Blue lines show linear regression fits, and grey bands represent 95% prediction confidence intervals. Spearman's correlation is indicated. (D) Distribution of ncORF PhyloCSF scores normalized by the number of codons in per ncORF. The dashed line denotes zero. CDS, coding sequence; ncORF, non-canonical open reading frame; ORF, open reading frame.

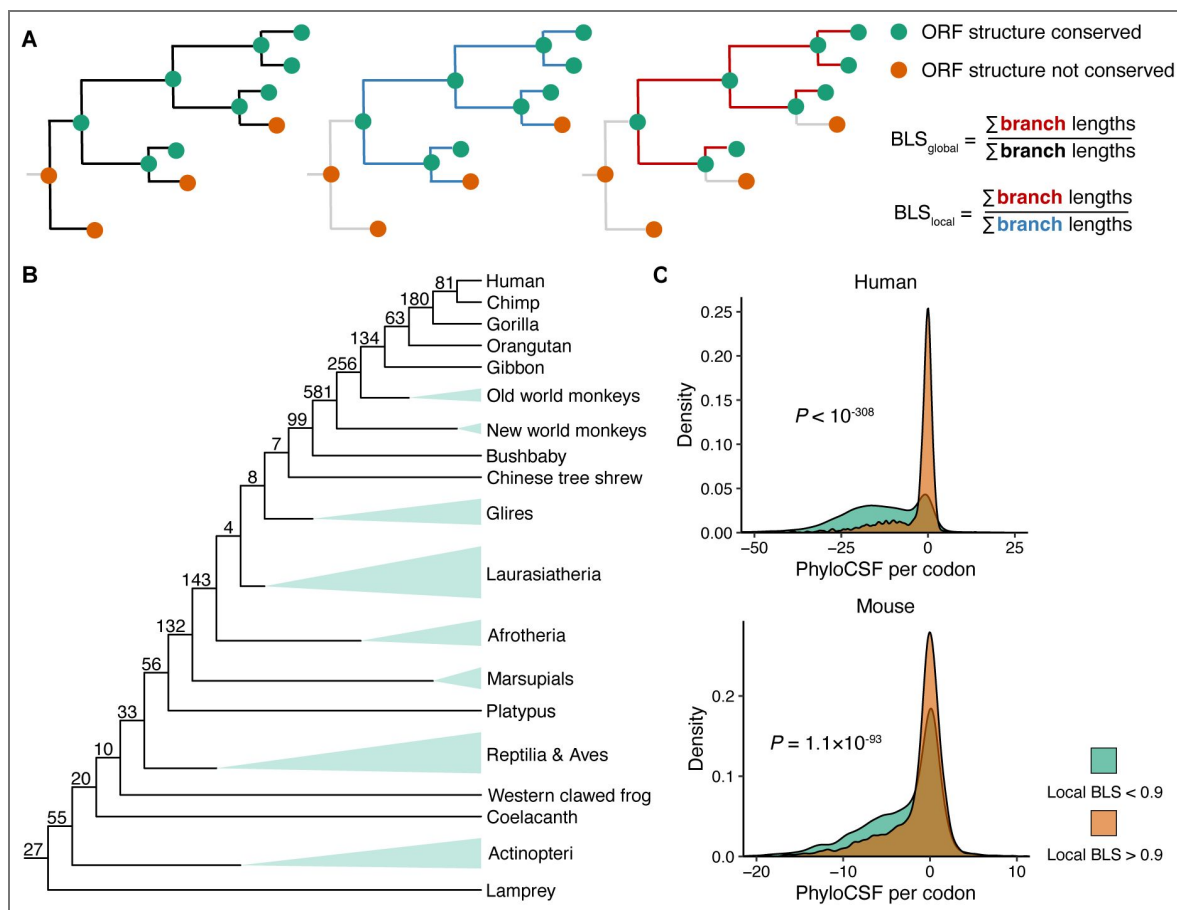


Figure 4. Lineage-specifically conserved ncORFs.

(A) Schematic illustration for BLS calculation. (B) Cladograms of vertebrates showing the number of lineage-specifically conserved ncORFs (local BLS > 0.9) at each ancestral node for humans and mice. (C) Distribution of PhyloCSF scores per codon for lineage-specifically conserved ncORFs compared to all other ncORFs. BLS, branch length score; ncORF, non-canonical open reading frame; ORF, open reading frame.

influence (Fig. S11B). To disentangle the effects of evolutionary constraint and gene age, we stratified ncORFs by origination nodes and examined their translation levels in relation to local BLS and PhyloCSF. Although ancient ncORFs are generally more highly translated, those with higher local BLS scores consistently show elevated translation levels compared to their age-matched counterparts (Figs. 5A & S12A). This suggests that lineage-specific conservation is associated with enhanced and possibly optimized translation, potentially reflecting acquired functional roles. Conversely, the relationship between coding potential and translation is more nuanced: young ncORFs with positive PhyloCSF scores tend to be more highly translated, while the opposite trend is observed in ancient ncORFs (Figs. 5B & S12B). These patterns point to complex interplays between evolutionary constraint, coding potential, and translational output, likely reflecting divergent selective pressures across different ncORF age groups.

While the above analyses revealed the overall trends in ncORF translation, they overlooked the heterogeneity across tissues and cell types. We found that ncORF translation exhibits strong tissue specificity (Fig. S13), which may stem from either transcriptional bias in ncORF-containing genes or post-transcriptional regulation specific to ncORFs. To disentangle these factors, we examined RNA levels of ncORF-containing genes across different tissues^{55,56}. Although transcriptional enrichment in testis was evident in both humans and mice (Fig. S14), testis-biased translation was observed only in mice (Fig. S13). Moreover, ncORF translation displayed significantly higher tissue specificity than the transcription of their host genes in both species (Fig. S15), indicating a substantial post-transcriptional component to their regulation. We next assessed how evolutionary characteristics shape tissue specificity. Ancient ncORFs tend to have lower tissue specificity than younger ones, suggesting they are more broadly translated (Figs. S16 & S12C). Intriguingly, coding potential exhibits opposite effects depending on evolutionary age: young ncORFs with higher coding potential are more broadly expressed, whereas ancient ones show increased tissue specificity (Figs. S17 & S12D). These patterns may help to explain the lower average translation levels observed for ancient ncORFs across tissues. A similar trend was also evident for ncORFs with high local constraint scores in mice (Fig. S12C). Collectively, these results support a model in which ncORFs initially evolve widespread expression to establish interactions and avoid loss⁵⁷, but may later become functionally specialized, resulting in more restricted expression.

Extensive co-translation between ncORFs and canonical CDSs

Given our earlier observation that many ncEPs lack modular domains and may act via interactions with canonical proteins, we next explored the landscape of co-translation between ncORFs and CDSs across different tissues or cell types as we expect that interacting proteins tend to be translated simultaneously spatiotemporally. Using weighted gene co-expression network analysis (WGCNA)⁵⁸, we identified 61,513 human and 55,449 mouse ncORF-CDS co-translated pairs. As expected, ncORFs in mRNAs tend to co-translate with the main CDS of the same gene, even after excluding CDS-overlapping ones (Fig. S15), consistent with previous experimental findings²⁸. This co-translation forms a highly modular bipartite network, in which one CDS may be linked to multiple ncORFs and vice versa (Fig. 6A-B). Network analysis revealed that hub CDS nodes—those with the most connections—in the largest modules are significantly enriched for functional gene categories (Fig. 6C-D). For instance, the largest module in humans is associated with chromatin-related proteins, while the second largest module in humans and the largest one in mice are linked to extracellular matrix organization. Notably, ancient ncORFs (those originating before the last common ancestor of mammals) are significantly more likely to co-translate with CDSs than younger ones (Fig. S18), potentially reflecting a greater likelihood of acquiring functional interactions over evolutionary time. In summary, our analyses revealed that evolutionary dynamics of ncORFs have widespread impacts on their translation and potential function.

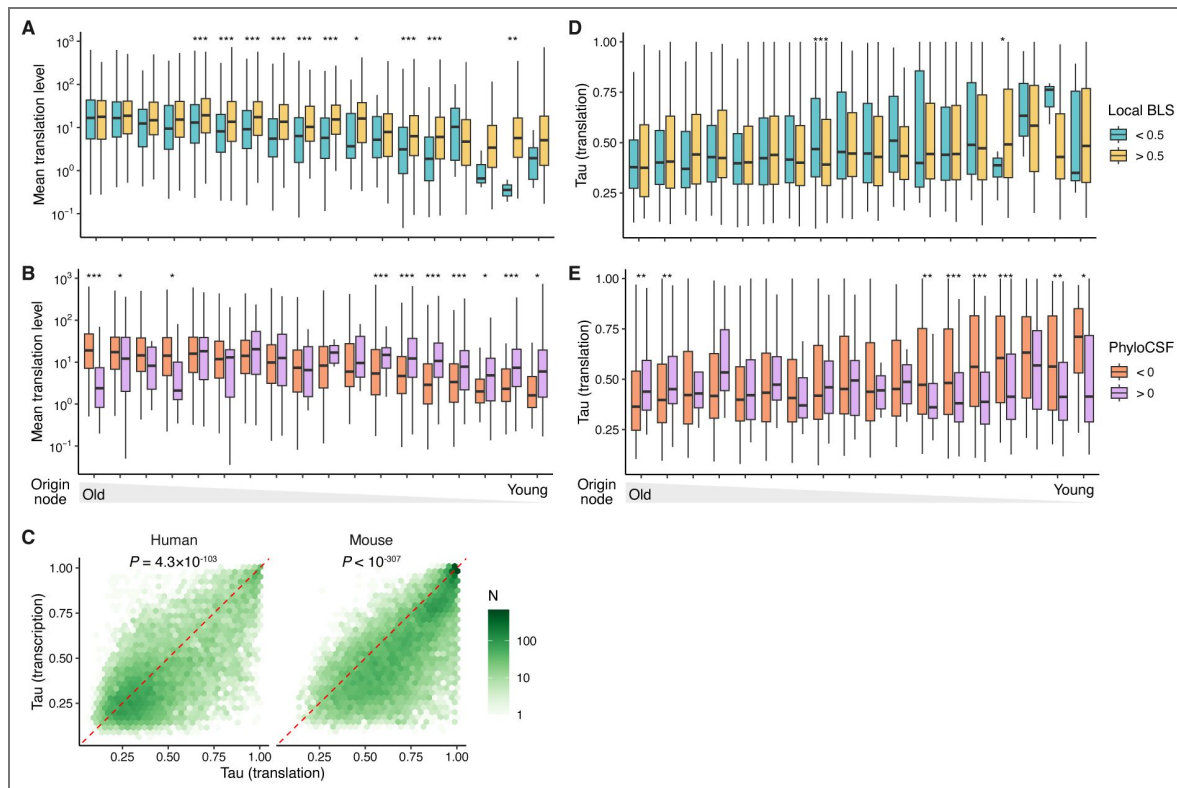


Figure 5. Evolutionary dynamics of ncORF expression.

(A) Distribution of mean translation levels of human ncORFs grouped by their origin nodes and further stratified by local BLS. Statistical significance was assessed using Wilcoxon rank-sum tests. ***, $P < 0.001$; **, $P < 0.01$; *, $P < 0.05$. (B) Similar to (A) but ncORFs are further stratified by local BLS. (C) Relationship between ncORF tissue specificity at translation and transcription levels. Differences were determined with Wilcoxon signed-rank tests. (D-E) Similar to (A) and (B) but showing the distribution of tissue specificity at the translation level. CDS, coding sequence; ncORF, non-canonical open reading frame; ORF, open reading frame.

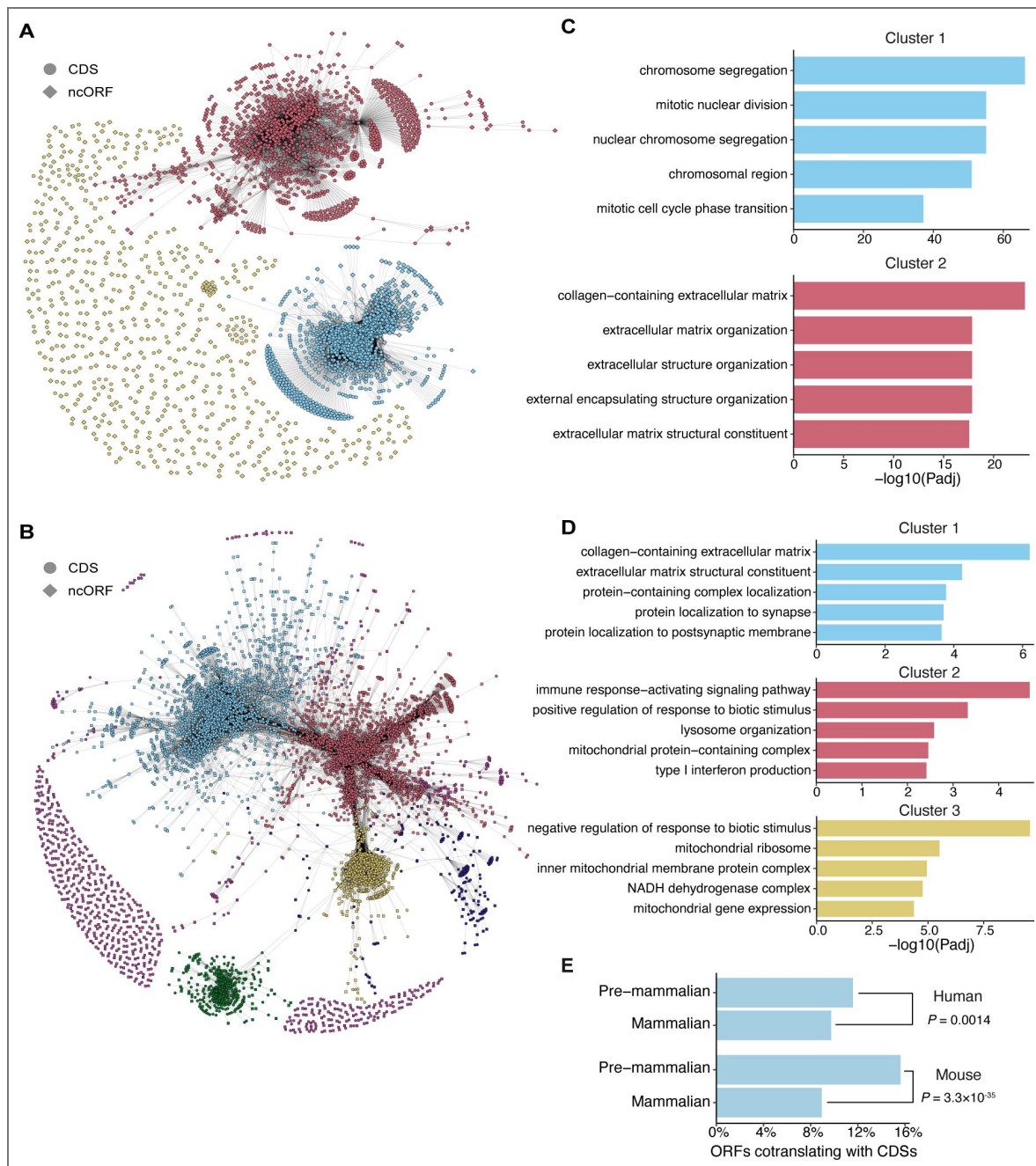


Figure 6. ncORF-CDS co-translation network.

(A) Bipartite network of ncORF-CDS co-translation in humans. (B) Bipartite network of ncORF-CDS co-translation in mice. ncORFs and CDSs are represented by different shapes and colored according to their cluster membership. Only the largest clusters were highlighted (top two in humans and top five in mice). (C) Top five enriched gene ontology terms for each of the two largest clusters in the human network shown in (A). (D) Top five enriched gene ontology terms for each of the three largest clusters in the mouse network shown in (B). (E) Proportion of ncORFs cotranslating with CDSs among ancient (pre-mammalian origin) versus younger (mammalian-specific) ncORFs. Differences were tested with Fisher's exact tests. CDS, coding sequence; ncORF, non-canonical open reading frame; ORF, open reading frame.

Discussion

Reliable annotation of translated ncORFs is critical for understanding their functional roles and evolutionary significance. In this study, we developed a standardized pipeline for the annotation of translated ncORFs and applied it to both humans and mice due to the extensive publicly available Ribo-Seq datasets. Compared to previous efforts, our annotation framework offers several key improvements. First, we selected only high-quality libraries with sufficient read coverage and acceptable in-frame RPF rates. Our earlier work demonstrated that both library size and in-frame rates impact the reproducibility and accuracy of ncORF predictions²⁰. This step helps mitigate false annotations arising from suboptimal data quality. Second, although many previously characterized functional ncEPs are associated with cancer-related ncORFs¹, ncORFs may also have important functions under physiological conditions. Importantly, such ncORFs are more likely to be subject to evolutionary constraints. Therefore, we restricted our analysis to normal cells and tissues, allowing us to identify more biologically relevant and conserved ncORFs. Third, given the variable performance across ncORF detection methods and the low reproducibility of predictions between biological replicates²⁰, we implemented stringent reproducibility criteria. Specifically, we retained only ncORFs detected by multiple methods and across independent datasets, enhancing the robustness of our final annotation. Finally, we applied a suite of well-validated filtering metrics from prior studies^{40–42} to minimize the inclusion of false positives, including those arising from translational noise^{59,60} or non-ribosomal contamination. As a result, our annotated ncORFs show significant enrichment for GENCODE ncORFs, especially those with MS evidence or reproducible detection across independent studies. These features collectively underscore the reliability and utility of our annotation for future functional and evolutionary studies of non-canonical translation.

From the perspective of new gene evolution, ncORFs represent a transitional stage between non-coding DNA/RNA and protein-coding genes^{61,62}. A major challenge in reclassifying ncORFs as bona fide coding genes lies in determining whether the putative proteins they encode are functionally relevant. Our analysis of ncEPs revealed that most lack recognizable protein domains and are enriched for IDRs. Therefore, we propose that many ncORFs may function in a non-autonomous manner by interacting with other proteins to exert their effects. This hypothesis aligns with prior studies showing physical interactions between ncEPs and known proteins^{28,30,35}. Supporting this view, our WGCNA analysis uncovered widespread co-translation between ncORFs and canonical CDSs, suggesting potential interaction partners and functional contexts.

We also assessed whether ncORFs are subject to evolutionary constraints consistent with functional protein coding. Using both population-level and cross-species analyses, we found that many ncORFs are under purifying selection, and their sequence conservation patterns often exhibit base-level features reminiscent of canonical CDSs—albeit to a lesser degree. Furthermore, we reconstructed the evolutionary trajectories of thousands of ncORFs in both humans and mice and observed sustained coding potential since their origination. Complementing these findings, we identified a subset of ncORFs that preserve intact ORF structures across most descendant species, suggesting ongoing purifying selection against disabling mutations. These lineage-specifically conserved and potentially coding ncORFs likely contribute to clade-specific functional innovation and represent compelling candidates for future functional characterization.

We further explored how evolutionary history shapes ncORF translation across tissues or cell types. Our results indicate that ancient ncORFs are more broadly and highly translated and have established more extensive co-expression with canonical CDSs. Moreover, when comparing ncORFs of similar age, those that are lineage-specifically constrained consistently exhibit higher translation levels, reinforcing their likely functional importance. Interestingly, we found that young ncORFs with strong evolutionary constraints tend to have lower tissue specificity in their translation, potentially reflecting a phase of widespread expression that facilitates functional recruitment. In contrast, older ncORFs with positive coding potential tend to display increased tissue specificity, implying functional specialization over evolutionary time. Collectively, these

findings support a model in which ncORFs evolve from broadly expressed, weakly functional elements into specialized, conserved effectors— though further empirical work is needed to validate this hypothesis.

In conclusion, we provide a comprehensive annotation of translated ncORFs in humans and mice, offering a valuable resource for functional and evolutionary genomics. Our integrated analysis of sequence evolution and translational landscape sheds light on the evolutionary trajectory and functional integration of ncORFs. The consistency of these patterns in both species suggests that our findings may generalize to other mammals. However, realizing the full biological significance of ncORFs, particularly lineage-specific ones, will require experimental validation. Recent advances in high-throughput screening technologies^{19,30,33,63} offer a promising path forward. Lastly, although our study focused on humans and mice due to the availability of high-quality Ribo-Seq data, we anticipate that our annotation will serve as a training resource for machine learning models aimed at predicting translated ncORFs in non-model organisms, thus broadening our understanding of genome coding potential across species.

Methods

Quality control and preprocessing of Ribo-Seq libraries

Ribo-Seq libraries from normal human or mouse samples without genetic modifications or perturbations (Table S7 [↗](#)) were downloaded from the European Nucleotide Archive (ENA) or the Short Read Archive (SRA). For ncORF annotation, libraries containing fewer than five million reads were excluded. Libraries derived from cancer tissues or cell lines were also removed based on sample metadata or the associated publications. Next, adapter sequences and any random nucleotide additions were trimmed using Cutadapt (v3.7)⁶⁴ and reads shorter than 18 nucleotides were discarded. Remaining reads were aligned to the human or mouse reference genome (GRCh38 or GRCm39, ENSEMBL⁶⁵ release 107) using the STAR aligner (v2.7.10a)⁶⁶. To assess library quality, the proportion of in-frame ribosome-protected fragment (RPF) reads was calculated using the PSite package⁶⁷. Libraries with ≥60% in-frame RPFs were classified as high quality. Only samples containing at least two high-quality libraries were retained for ncORF annotation (Table S1 [↗](#)). For the expression analyses, all the downloaded Ribo-Seq libraries were used to cover more samples.

ncORF annotation

For each Ribo-Seq library, translated ORFs initiating with AUG, CUG, GUG, or UUG and encoding at least five amino acids were predicted using PRICE³⁷, RiboCode³⁸, and Ribo-TISH³⁹, as described previously²⁰. To define novel ORFs, annotated CDSs and their truncation or extension variants (i.e., ORFs with the same start or stop position as annotated CDSs) were excluded. To eliminate likely false positives resulting from non-ribosomal contamination or stochastic molecular errors, we calculated RRS^{41,42} and FLOSS⁴⁰ of each ncORF. RRS measures the drop in RPF coverage after stop codon—a key signature of active translation beyond three-nucleotide periodicity^{41,42}, whereas FLOSS assesses whether the RPF length distribution resembles that of canonical CDSs, helping to distinguish authentic ORFs from non-ribosomal signals⁴⁰. In each library, uniquely mapped RPFs were assigned to the corresponding P-site, and genome-wide P-site coverage was computed with the PSite package. Similar to previous studies^{41,42}, RRS was calculated as the sum of coverage across the five codons preceding the stop codon divided by the sum across the five trinucleotides following the stop codon. FLOSS was calculated by first deriving a reference RPF length distribution vector (P_{ref}) using all RPFs uniquely mapped to annotated CDSs. For each ORF (ncORF or CDS), a read length frequency vector (P_{ORF}) was computed, and FLOSS was calculated as $\sum |P_{ref} - P_{ORF}|/2$. The empirical upper bound for acceptable FLOSS was set using Tukey's method (i.e., the third quartile + $1.5 \times$ inter-quartile range of FLOSS values across all CDSs)⁴⁰. ncORFs with fewer than 10 P-site-mapped RPFs, $RRS \leq 1$, or FLOSS values exceeding the empirical upper bound were filtered out.

Integration of ncORF prediction across different libraries of the same species was nontrivial due to extensive overlap among ncORFs in genomic coordinates. To address this issue, we used a graph-based clustering approach. We created a graph where two ORF vertices were connected if they share the same start or stop codon. Then, each set of connected ncORFs in the graph represents a cluster of overlapping ncORFs. To ensure reproducibility, only clusters identified by ≥ 2 methods in ≥ 2 libraries were retained, yielding 13,071 clusters in humans and 17,674 clusters in mice. To ensure compatibility with the latest reference gene annotations (ENSEMBL release 111), we removed any ncORFs that were newly classified as annotated CDSs. A manual inspection of RPF signals using the GWIPS-viz Ribo-Seq track⁶⁸ in the UCSC Genome Browser⁶⁹ revealed additional spurious ncORFs overlapping in-frame with CDSs from other transcripts. These were likely misidentified due to shared translation signals and were conservatively excluded using custom scripts, leaving 11,623 clusters in humans and 16,485 in mice.

To facilitate downstream analyses, a representative ncORF was selected for each cluster using a two-step strategy. First, a representative transcript was selected by prioritizing those with matched annotation from NCBI and EMBL-EBI (MANE)⁴³ (for humans) or APPRIS principal⁷⁰; otherwise, the longest transcript was chosen. Then, among ncORFs within the representative transcript, a single ncORF was selected based on start codon hierarchy (AUG > CUG/GUG > UUG) following empirical evidence^{28,37}. If multiple ncORFs shared the same preferred start codon, the longer one was chosen.

For validation, our ncORFs were compared with previously annotated GENCODE ncORFs² and those with MS evidence compiled previously²⁰. Tier annotations of GENCODE ncORFs were obtained from a recent study²⁹. Known functional ncORFs and their genomic coordinates were manually curated from published literature (Table S4 [4](#)). To account for possible proteoforms arising from alternative initiation and splicing, we considered an ncORF to be recovered if either its start or stop codon was identical to any ncORF in our annotation.

ncORF sequence analyses

The secondary structure surrounding each ORF (ncORF or CDS) start codon was predicted with RNAfold⁷¹ on a 150 nt window extracted from the transcript in which the ncORF resides starting from the 30th nucleotide upstream of the start codon. For each position within this window, the overall pairing probability was calculated by dividing the number of ORFs with a base-paired residue at that position by the total number of ORFs analyzed. The tRNA adaptation index and effective number of codons for ncORFs and CDSs were calculated using the “cubar” R package⁷². tRNA gene annotations in humans and mice were obtained from GtRNAdb (release 22)⁷³.

Putative ncEPs analyses

Known domains within ncEPs were identified using InterProScan (v5.64-96.0)⁷⁴ with default parameters. Repetitive sequence annotations were downloaded from the UCSC Table Browser. BEDTools (v2.31.1)⁷⁵ was used to identify TE-overlapping ncORFs. IDRs were predicted with IUPred3⁷⁶ with analysis type set to “short disorders” and smoothing disabled. For each ORF, residues with disorder tendency score > 0.5 were considered disordered, and the IDR fraction was calculated as the proportion of disordered residues relative to the ORF length.

Gnocchi Score and PhyloP of ncORFs and CDSs

Genome-wide Gnocchi scores computed using 1 kb sliding windows (with 100bp step size) were downloaded from the gnomAD database⁵⁰. For each base pair, the Gnocchi score was defined as the average across the ten overlapping 1kb windows that covered that position using BEDTools. The final Gnocchi score for each ORF (ncORF or CDS) was obtained by averaging the scores of all base pairs within the ORF.

Base-wise PhyloP scores computed across 441 mammals or 239 primates⁵¹ were downloaded from the UCSC Genome Browser. For each ncORF or CDS, PhyloP values were extracted for a ± 30 bp window surrounding the start and stop codons using custom scripts. These values were then averaged over different ORFs to generate metagene profiles for visualization. To evaluate three-

nucleotide periodicity, the metagene profiles were subjected to autocorrelation analysis using the “ac.test” function from the “testcorr” R package⁷⁷. Statistical significance was assessed using the robust $\tilde{t}$ statistics.

ncORF origination node and mode

Whole genome alignments and corresponding species trees were downloaded from the UCSC Genome Browser (GRCh38 100-way for humans and GRCm39 35-way for mice). The origination node and mode for each ncORF were inferred following previous approaches^{34,35}. For each ORF, the sequence from the reference genome and its orthologous regions in other species were extracted. Orthologous sequences covering less than 50% of the ncORF length relative to the reference species were excluded. The remaining sequences were realigned with MAFFT (v7.520)⁷⁸ with default parameters. The corresponding gene tree was derived by pruning the species tree using Gtree (v0.4.3)⁷⁹ with default settings. Ancestral sequence reconstruction was performed using FastML (v3.11)⁸⁰ with parameters “-- seqType nuc --SubMatrix HKY --OptimizeBL no --indelReconstruction ML”. An ORF structure was considered intact in an orthologous or ancestral sequence if one of the first three codons was a start codon and at least 70% of the 5' region of the reference ncORF was free of in-frame stop codons. The origination node was defined as the most ancient ancestral node that retained an intact ORF structure. An ncORF was classified as *de novo* if the origination node had an ancestor covering at least 50% of the ncORF sequence but lacking an intact ncORF structure. Otherwise, the origination mode was classified as uncertain. The distance from an ancestral node to the reference species (tip node) was calculated by summing branch lengths along the connecting path using the ape R package (v5.8)⁸¹.

ncORF BLS and PhyloCSF

For each ncORF, global BLS was calculated as the ratio of the total branch length of the subtree connecting all leaves with intact ORF structures to the total branch length of the full species tree as in our previous study⁴⁸ with custom scripts. Local BLS was calculated similarly, but the denominator was replaced by the total branch length of the subtree defined by the origination node and all of its descendant nodes (including leaves). To evaluate coding potential, the alignment of all species descending from the origination node was converted into Multiple Alignment Format after removing the stop codon. PhyloCSF scores were then computed using the “score-msa” module of PhyloCSF++⁸².

ncORF expression analysis

To quantify translation levels of ncORFs and CDSs in Ribo-Seq libraries, we assigned uniquely mapped reads to corresponding P-site positions with the PSite package. The read count for each ORF was calculated as the total number of RPFs whose P-sites fell within the ORF. To reduce bias caused by RPF enrichment at ORF termini, reads mapping to start or stop codons were excluded. For CDSs, only the count of the representative transcript of each gene as defined above was used for downstream analyses. For samples with multiple libraries, read counts were aggregated prior to normalization. Sample-specific size factors were estimated using CDS counts and applied to normalize both ncORF and CDS counts across samples using DESeq2⁸³. RPKM for an ORF was calculated as normalized read counts / ORF length (nt) / library size $\times 10^9$, where library size was the median of total read counts across samples to ensure cross-sample comparability. An ORF (ncORF or CDS) was considered translated if its RPKM was ≥ 1 . To evaluate tissue specificity of ncORF translation, the normalized RPF count X_k of an ORF in sample k was log-transformed as $X_k = \log(X_k + 1)$. Then, tissue specificity index⁸⁴ was calculated as $\tau = \sum_k^n (1 - \tilde{x}_k) / (n - 1)$, where $\tilde{x}_k = x_k / \max_{1 \leq j \leq n} x_j$ and n is the total number of samples. To quantify RNA levels, raw reads of deep transcriptome profiling across different human or mouse tissues from previous studies^{55,56} were downloaded from SRA. Gene expression levels (transcript per million, TPM) were estimated using salmon (v1.10.3)⁸⁵ with parameters “--seqBias --gcBias --posBias -l A”.

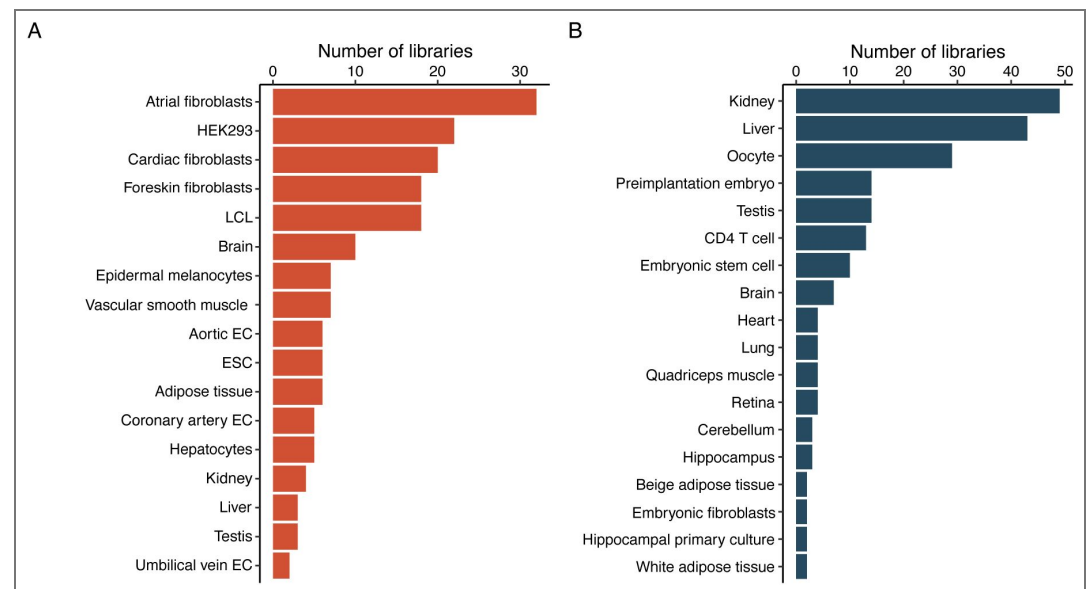
WGCNA analysis⁵⁸ of ncORF and CDS translation was performed using the PyWGCNA package (v2.0.5)⁸⁶ with default settings. Only ORFs with normalized RPF counts ≥ 10 in more than half of the samples were included. The soft threshold was determined automatically by PyWGCNA. Due to differences in soft thresholds, two ORFs were considered cotranslated if their topological overlap > 0.1 (human) or > 0.01 (mouse).⁵⁸ To test whether co-translation is enriched between ncORFs and CDSs from the same transcript, gene identities of CDSs were randomly permuted among all co-translated ncORF–CDS pairs. This procedure was repeated 1,000 times, and the empirical P value was calculated as $(m + 1) / 1,000$, where m is the number of permutations in which the number of same-gene pairs was greater than or equal to the observed count. The bipartite network of ncORF–CDS co-translation was clustered using the Leiden algorithm⁸⁷ implemented in igraph (v2.1.3)⁸⁸ by optimizing modularity with a resolution of 0.2. The top 20% of nodes by betweenness centrality were designated as hub nodes. Gene ontology analysis of hub genes in the largest modules was conducted using ClusterProfiler (v4.12)⁸⁹.

Data availability

Ribo-Seq libraries for ncORF annotation were obtained from the ENA or SRA, with accession codes listed in Table S1⁸⁴. Libraries used for ncORF and CDS translation quantification were listed in Table S7⁸⁴. RNA-Seq libraries for gene expression profiling across different human or mouse tissues were from previous studies^{55,56} and are available with ENA accession numbers E-MTAB-2836 and E-MTAB-10276, respectively. Whole genome alignment across vertebrates, annotation of genomic repetitive elements, and base-wise PhyloP in primates and mammals⁵¹ were downloaded from the UCSC Genome Browser⁶⁹. Gnocchi scores were downloaded from gnomAD⁵⁰. All the other data were presented in the manuscript or supplementary tables and are available from the corresponding author upon reasonable request.

Supplementary Information

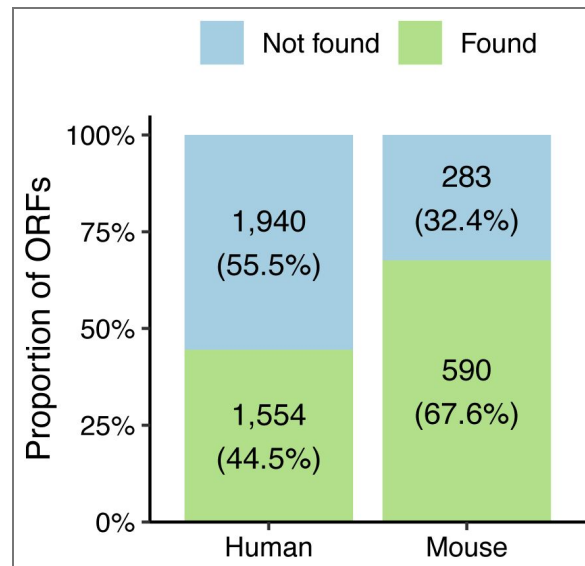
Supplementary Figures



Supplementary figure 1. Number of high-quality Ribo-Seq libraries passing quality control for each sample in humans (A) and mice (B). EC, endothelial cell; ESC, embryonic stem cell; LCL, lymphoblastoid cell line; Ribo-Seq, ribosome profiling.

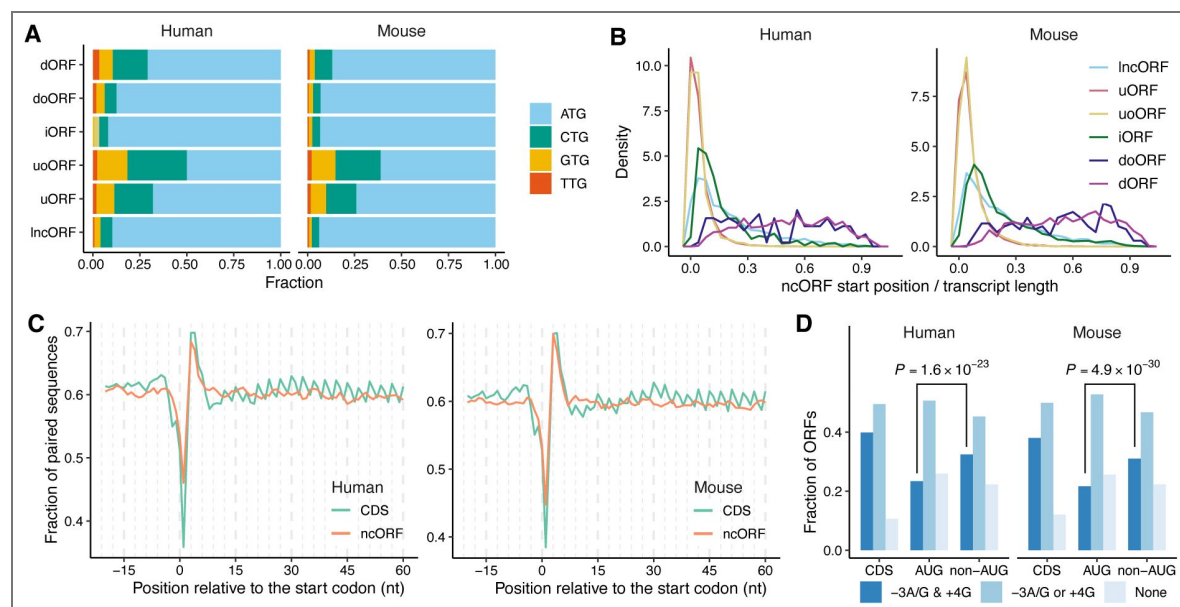
Supplementary figure 2. Rediscovery rate of non-canonical ORFs supported by MS evidence, as compiled in our previous study²⁰.

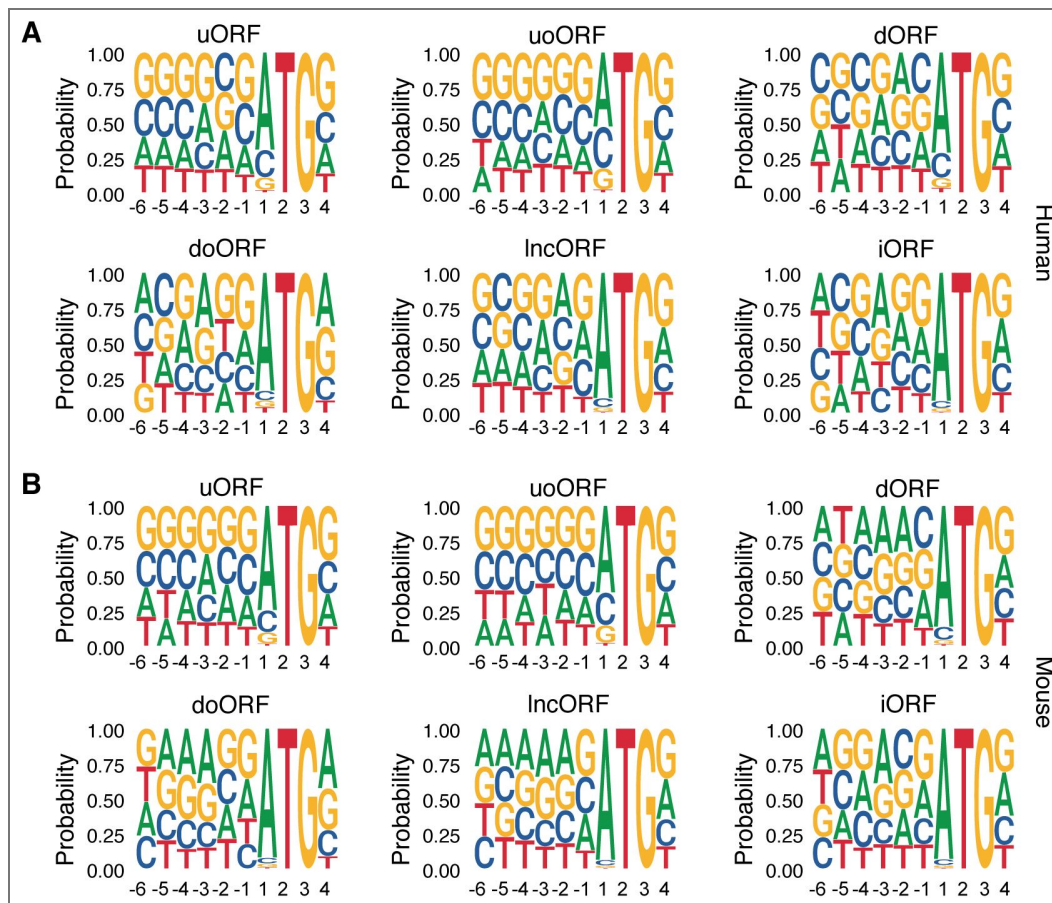
MS, mass spectrometry; ORF, open reading frame.



Supplementary figure 3. Translation-related signatures of ncORF sequences.

(A) Start codon usage among ncORFs. (B) Distribution of relative start positions of ncORFs within transcripts. The x-axis represents the position of ncORF start codon normalized by transcript length. (C) Nucleotide pairing probabilities around the start codons of ncORFs and CDSs. (D) Proportion of ncORFs with different Kozak sequence contexts. Statistical significance was assessed using Fisher's exact test. CDS, coding sequence; ncORF, non-canonical open reading frame; ORF, open reading frame.



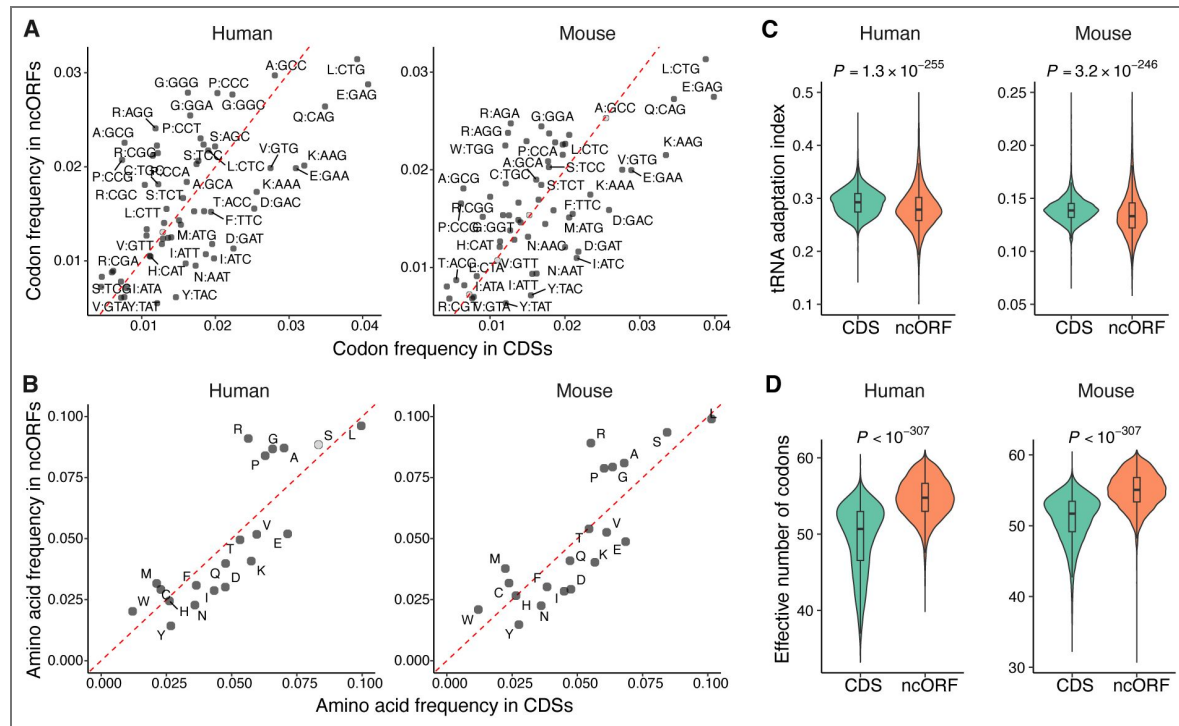


Supplementary figure 4. Seqlogo plots of Kozak sequence context surrounding ncORF start codons in humans (A) and mice (B).

The first nucleotide of the start codon is designated as position +1. The height of each nucleotide at each position reflects its frequency. ncORF, non-canonical open reading frame.

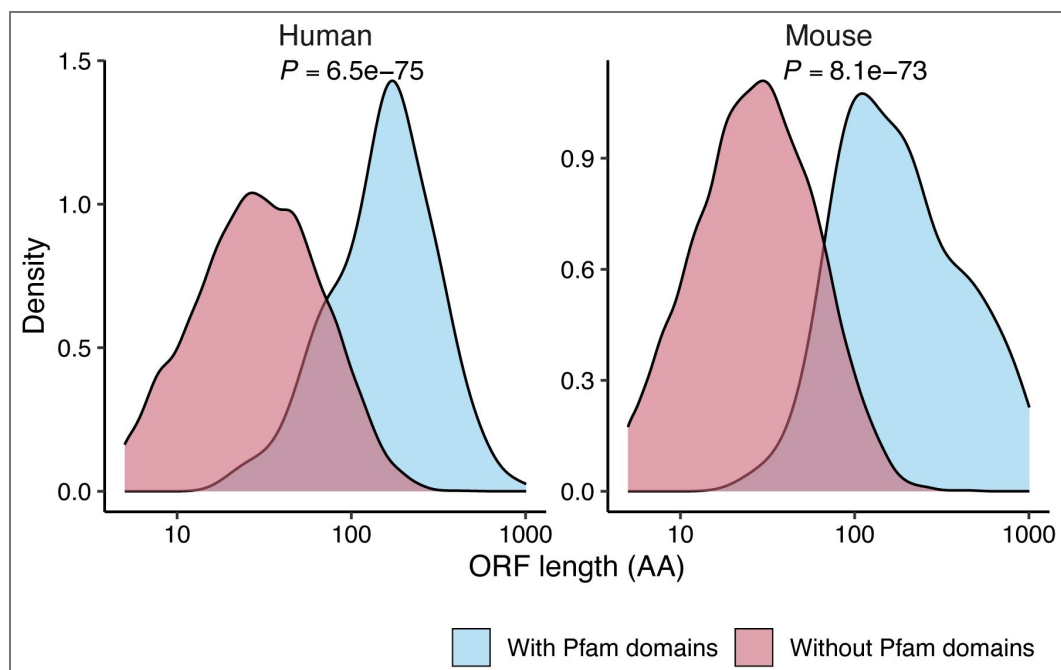
Supplementary figure 5. Differences in codon and amino acid usage between ncORFs and CDSs.

(A-B) Differences in relative usage of codons (A) and amino acids (B). Significance was determined using Fisher's exact tests and corrected for multiple-testing. Codons or amino acids with adjusted P value < 0.05 are shown in dark grey. (C-D) Distribution of tRNA adaptation index values (C) and effective number of codons (D) for CDSs and ncORFs. Statistical significance was determined by Wilcoxon rank-sum tests. CDS, coding sequence; ncORF, non-canonical open reading frame.



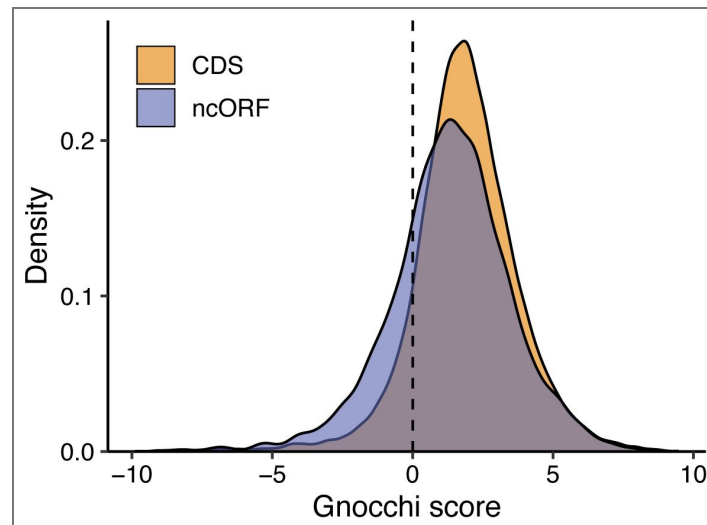
Supplementary figure 6. Length distribution of ncORFs with (blue) or without (red) known Pfam domains.

Statistical differences were assessed using Wilcoxon rank-sum tests. AA, amino acid; ncORF, non-canonical open reading frame; ORF, open reading frame.



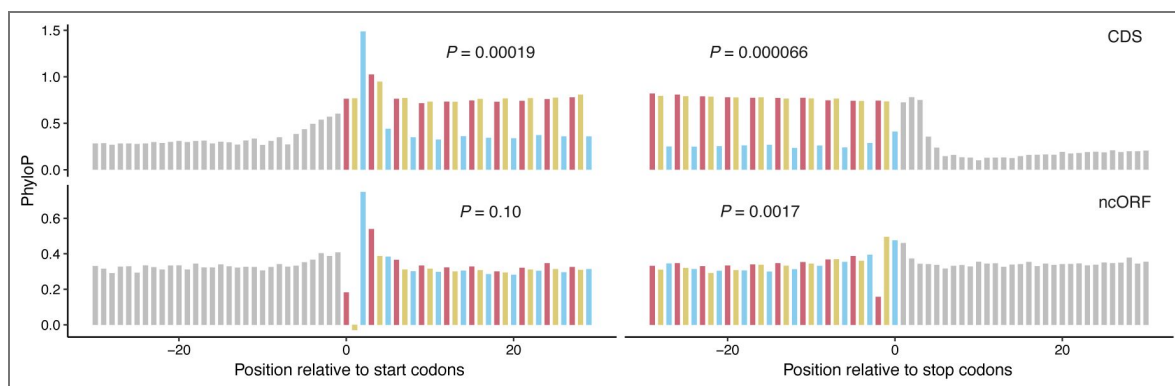
Supplementary figure 7. Distribution of Gnocchi scores for CDSs and ncORFs.

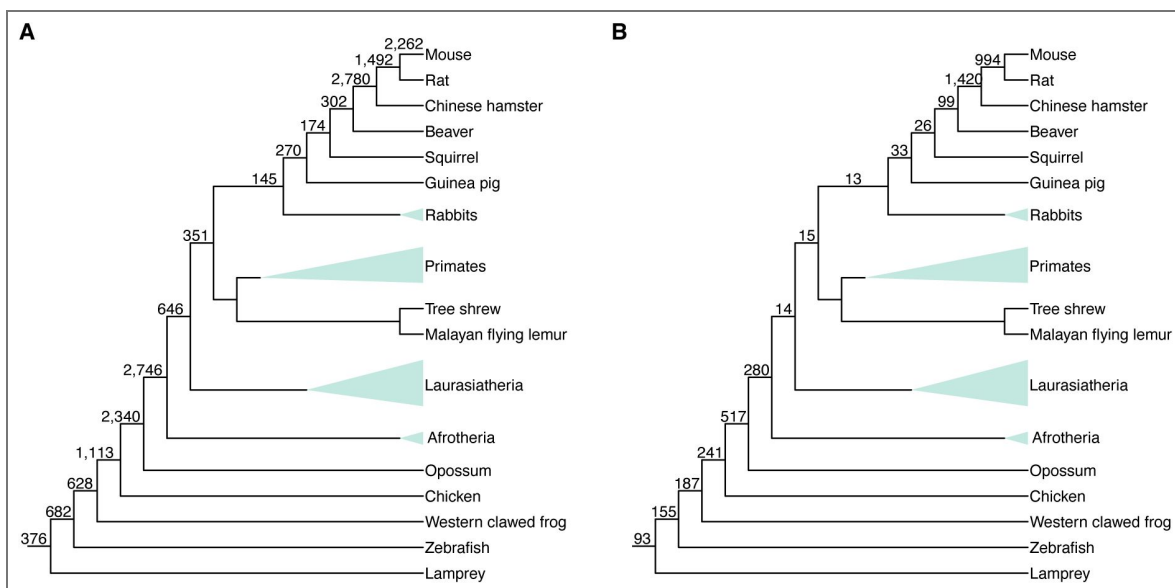
Statistical significance of deviation from zero was assessed using Wilcoxon signed-rank tests. CDS, coding sequence; Gnocchi, Genomic Non-Coding Constraint of Haplo-Insufficient variation; ncORF, non-canonical open reading frame.



Supplementary figure 8. The mean PhyloP score of 30 base pairs upstream and downstream of the ORF (CDS or ncORF) start or stop codons in primates.

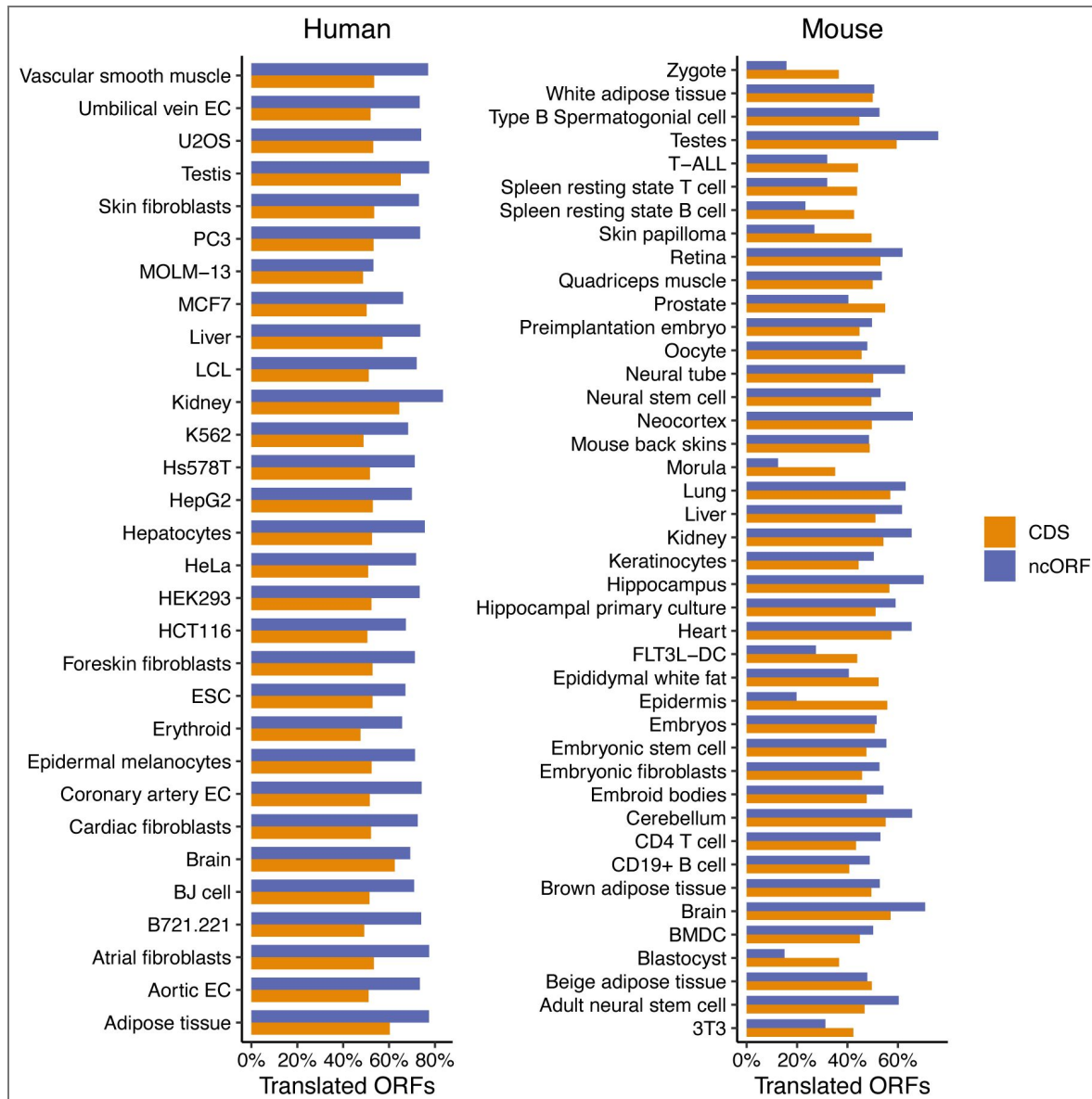
Codons were delineated with bars of alternating colors for different frames, and nucleotides in untranslated regions were shown in grey. Statistical significance of the three-nucleotide periodicity was determined by autocorrelation with a lag of three. CDS, coding sequence; ncORF, non-canonical open reading frame.





Supplementary figure 9. Cladograms of vertebrates showing the number of ncORFs originating at each ancestral branch leading to mice.

(A) All ncORFs. (B) Lineage-specific ncORFs with local branch length score > 0.9. Triangles indicate species merged into larger clades for visual simplicity.

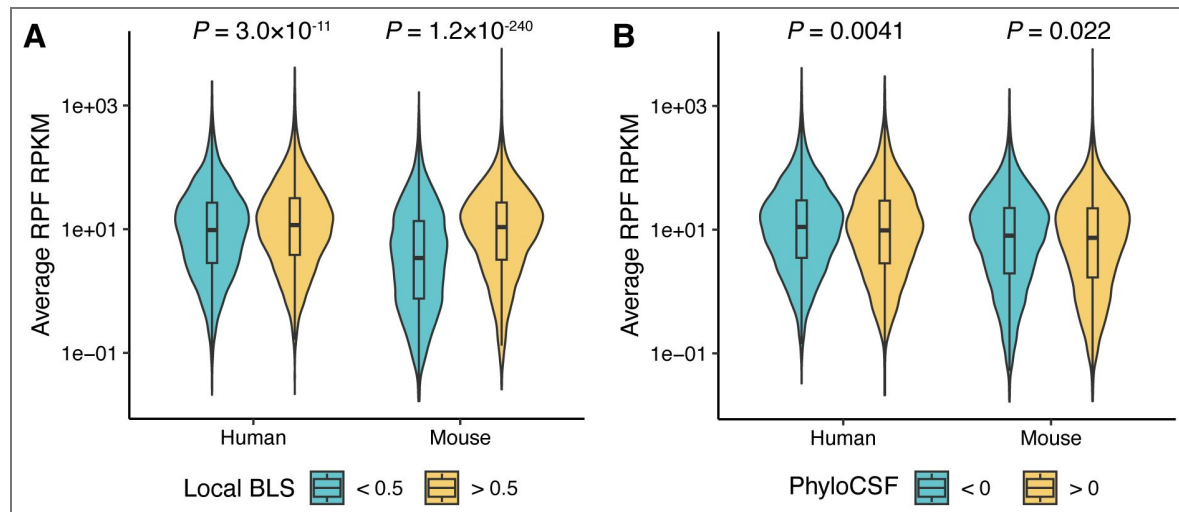


Supplementary figure 10. Proportion of translated ncORFs and CDSs detected in each human and mouse sample.

Translated ORFs were detected with RPKM ≥ 1 . BMDC, bone marrow-derived dendritic cell; CDS, coding sequence; DC, dendritic cell; EC, endothelial cell; ESC, embryonic stem cell; LCL, lymphoblastoid cell line; ncORF, non-canonical open reading frame; RPKM, reads per kilobase per million mapped reads; T-ALL, T-cell acute lymphoblastic leukemia.

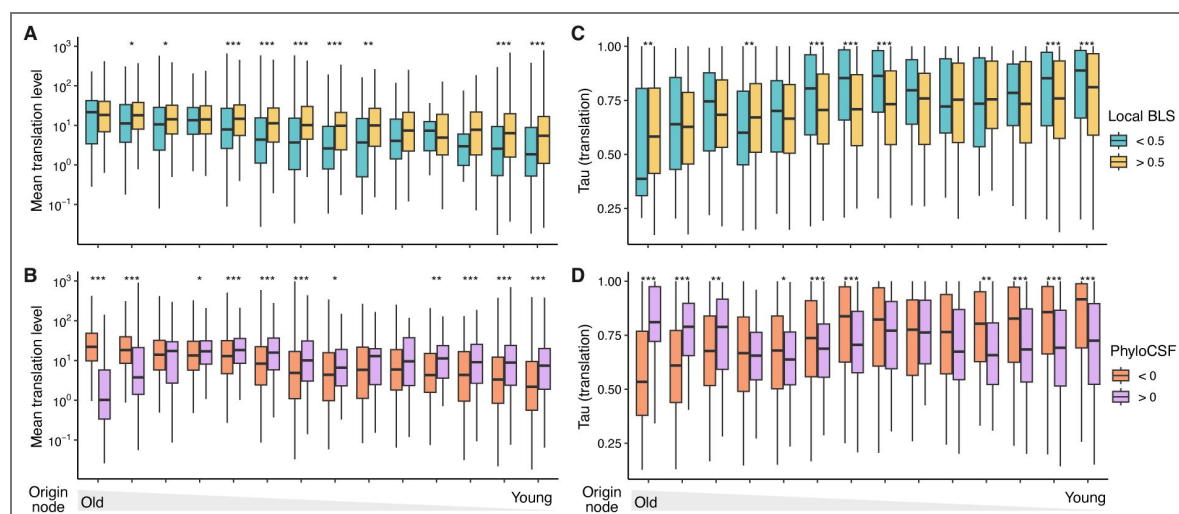
Supplementary figure 11. Distribution of average translation levels of ncORFs stratified by local BLS (A) or PhyloCSF (B) in humans and mice.

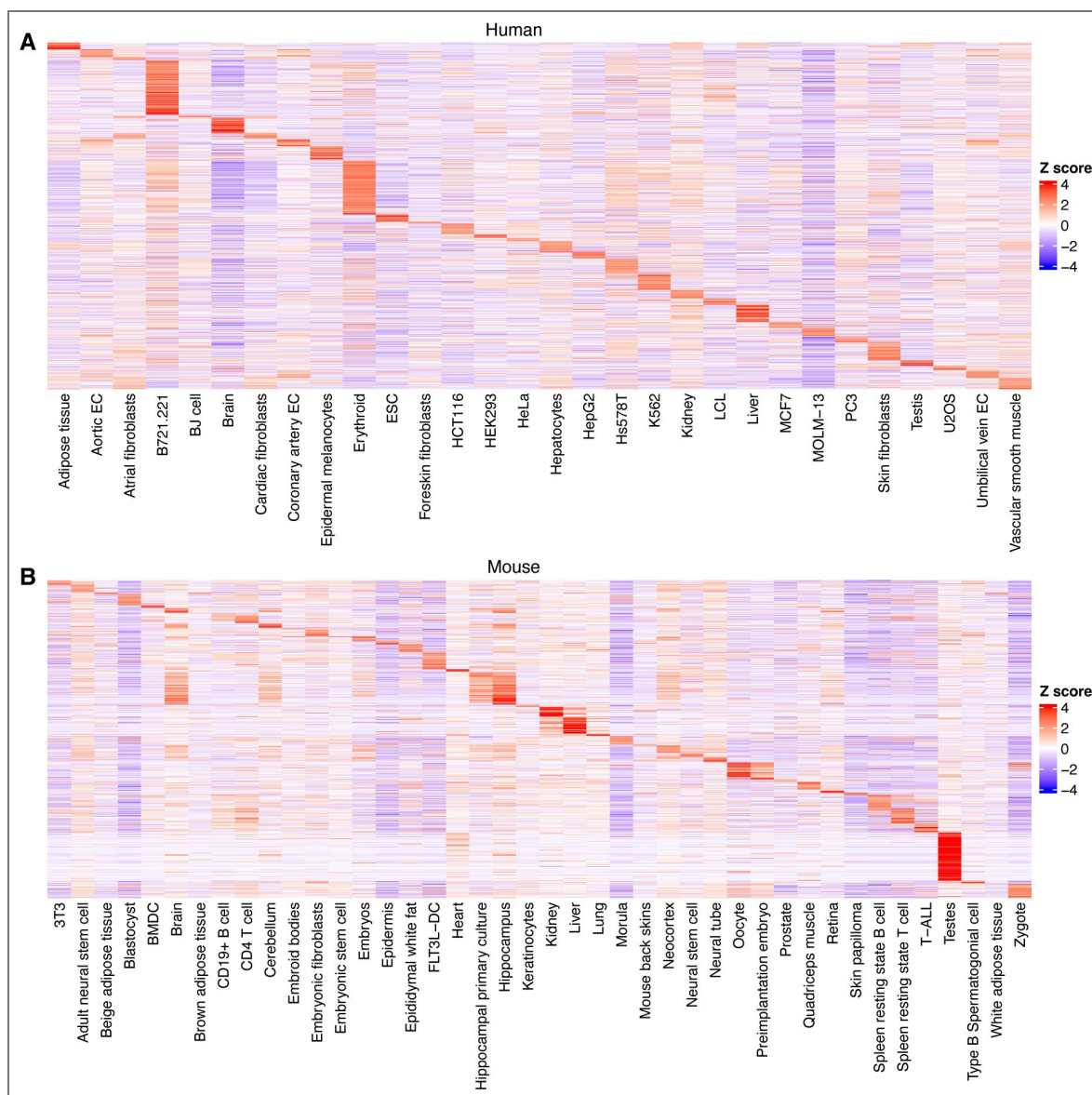
Statistical differences were assessed using Wilcoxon rank-sum tests. BLS, branch length score; ncORF, non-canonical open reading frame; RPF, ribosome-protected fragment; RPKM, reads per kilobase per million mapped reads.



Supplementary figure 12. Expression patterns of ncORFs in mice.

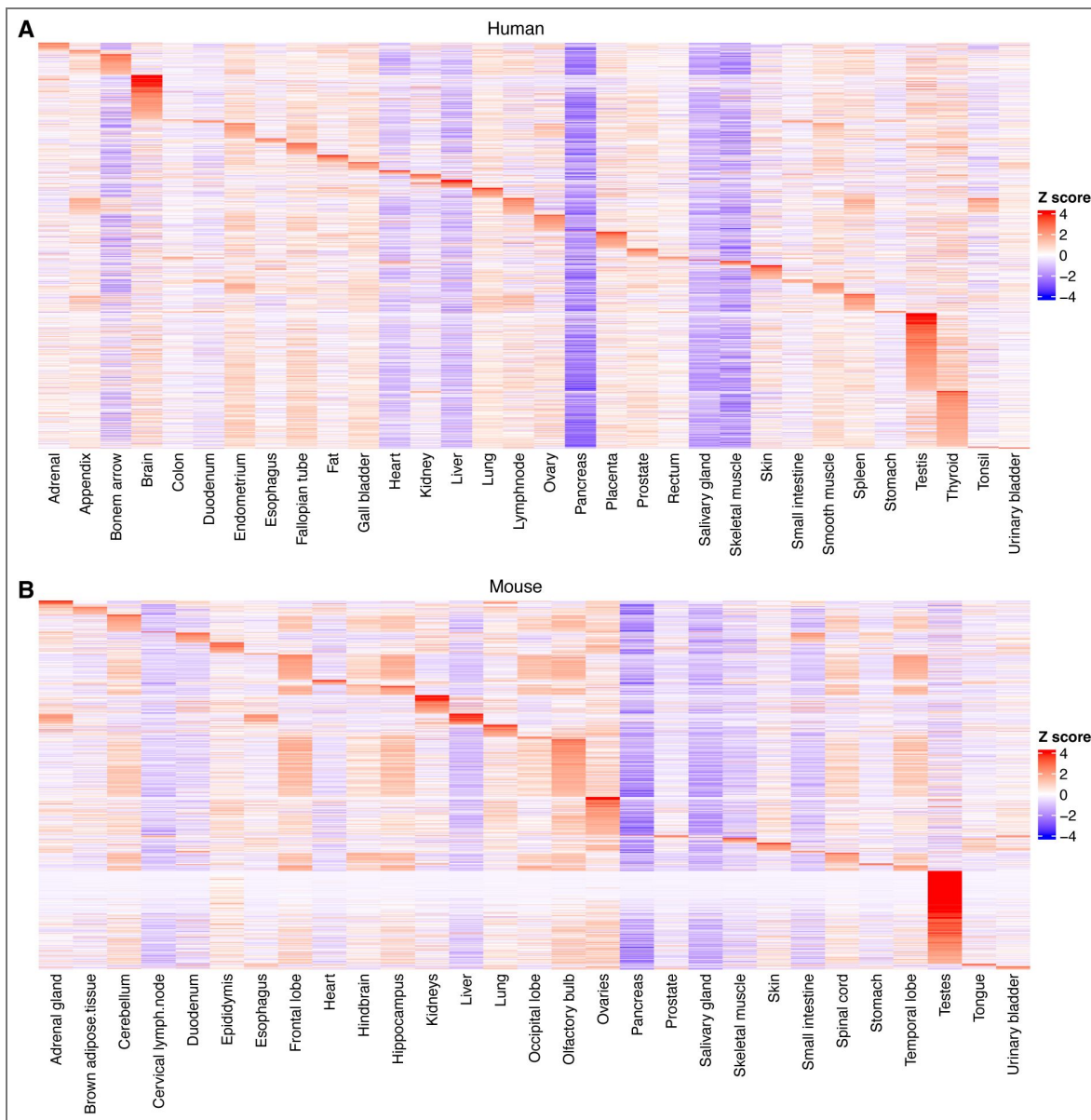
(A) Distribution of mean translation levels of mouse ncORFs grouped by origin nodes and further stratified by local BLS. Statistical significance was assessed with Wilcoxon rank-sum tests. ***, $P < 0.001$; **, $P < 0.01$; *, $P < 0.05$. (B) Similar to (A) but ncORFs are further stratified by PhyloCSF scores. (C-D) Similar to (A) and (B), but showing the distribution of tissue-specificity at the translation level.





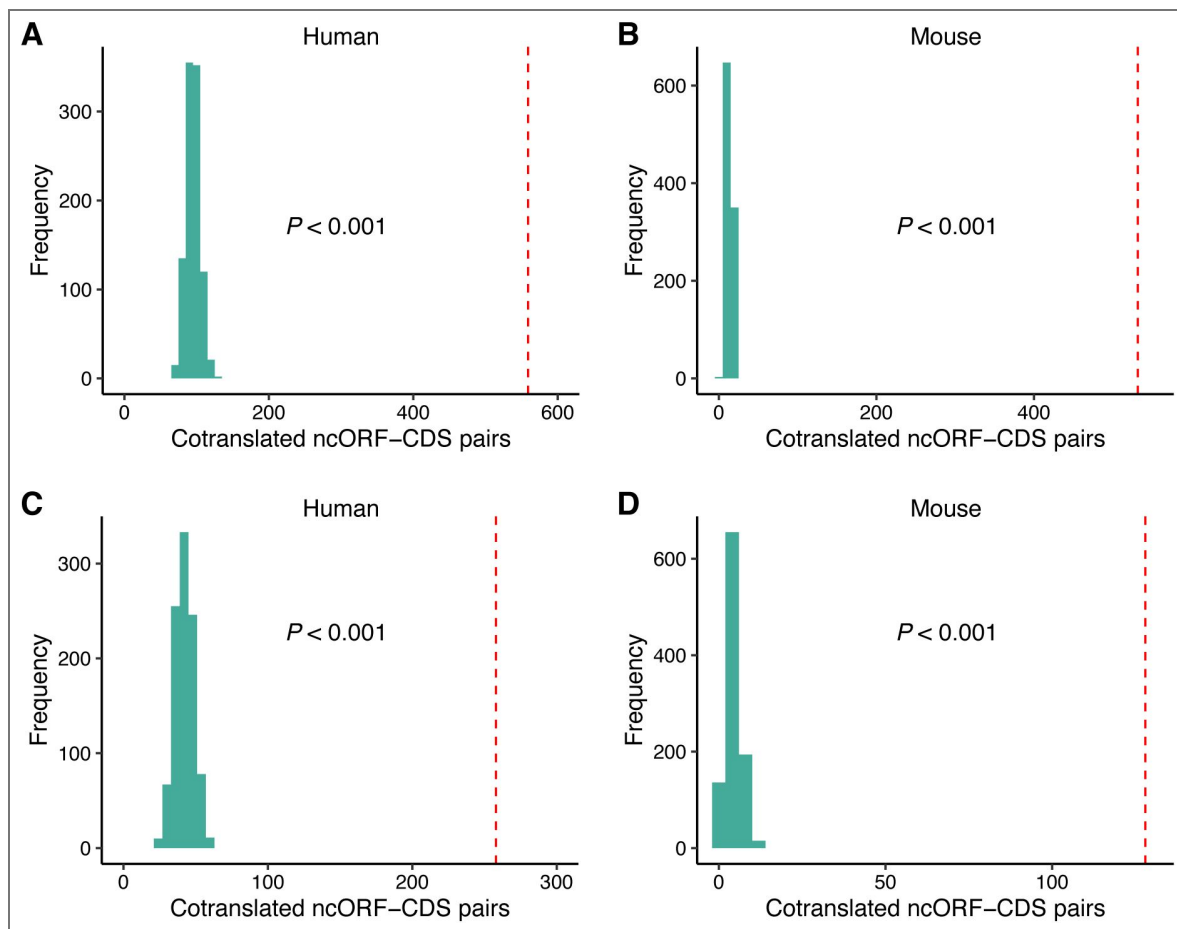
Supplementary figure 13. Heatmap showing the translation levels of ncORFs in human (A) and mouse (B) samples.

Only ncORFs with translation level (RPKM) ≥ 1 in at least one sample are displayed. Translation levels were normalized across samples by Z-score transformation before visualization. BMDC, bone marrow-derived dendritic cell; DC, dendritic cell; EC, endothelial cell; ESC, embryonic stem cell; LCL, lymphoblastoid cell line; ncORF, non-canonical open reading frame; Ribo-Seq, ribosome profiling; RPKM, reads per kilobase per million mapped reads; T-ALL, T-cell acute lymphoblastic leukemia.



Supplementary figure 14. Heatmap showing the transcription levels of ncORF-containing genes in humans (A) and mice (B).

Only genes expressed with TPM ≥ 1 in at least one sample are shown. TPM values were normalized across samples by Z-score transformation prior to visualization. ncORF, non-canonical open reading frame; TPM, transcripts per million.



Supplementary figure 15. Permutation tests assessing co-translation between ncORFs and main CDSs within the same genes.

(A-B) Distribution of the number of ncORFs that co-translate with corresponding main CDSs of the same genes in humans (A) and mice (B) based on the permutation analysis. Red dashed lines mark the observed counts. Empirical P values were indicated. (C-D) Same analyses as in (A) and (B), but excluding ncORFs overlapping with any CDS regions. CDS, coding sequence; ncORF, non-canonical open reading frame.

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Additional information

Code availability

All analyses were primarily conducted using R software (v4.3). Unless otherwise stated, statistical tests are two-sided. The code is available at https://github.com/gxlab/ncorf_mammals and custom scripts can be found at <https://github.com/gxlab/scripts>.

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Additional files

Supplementary Tables [Supplementary table 1](#). High-quality ribosome profiling libraries used for non-canonical open reading frame annotation. **Supplementary table 2**. Clusters of genomically overlapping GENCODE non-canonical open reading frames. **Supplementary table 3**. List of non-canonical open reading frames annotated in this study. **Supplementary table 4**. Known non-canonical open reading frames previously reported to encode functional microproteins and independently rediscovered here. **Supplementary table 5**. Non-canonical open reading frames containing Pfam domains. **Supplementary table 6**. Evolutionary characteristics and constraints of non-canonical open reading frames. **Supplementary table 7**. List of ribosome profiling libraries used for translation quantification.

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Peer reviews

Reviewer #1 (Public review):

This work compiles a comprehensive atlas of ncORFs across mammalian tissues and cell types, derived from reanalysis of ~400 public ribosome profiling datasets. The authors then evaluate cross-species conservation and functional signatures, proposing that evolutionarily ancient ncORFs tend to have higher translation potential, stronger expression, and closer relationships with canonical coding sequences.

Strengths:

In general, the study provides a large-scale and timely resource of annotated ncORFs, which could be broadly useful for the community. The authors collected ~400 public ribosome profiling datasets for annotations of ncORFs, which, to my best knowledge, is the largest collection of data for such a purpose. The catalog could facilitate future investigations into ncORF biology and broaden understanding of the coding potential of the "non-coding" genome.

Weaknesses:

Based on the ncORF catalog, some of the analyses were not properly done. Some of the results are descriptive.

- (1) Bias and representations of the data source. Public ribo-seq datasets are unevenly distributed across tissues and cell lines, raising concerns about heterogeneity and underrepresentation of certain contexts. This may limit the generalizability of the catalog.
- (2) The discussion on modular domains of ncORFs is unclear, and the claim that they may originate via TE-related mechanisms is not well supported. Stronger evidence or clearer reasoning is needed.
- (3) The conservation comparisons are not fully convincing. Figure S7 shows only mild differences between ncORFs and CDS, and statistical significance is not clearly demonstrated. Comparisons with other non-coding RNAs should be added, and overlapping sequences between ncORFs and CDS should be excluded to avoid bias.
- (4) Figure 3 indicates that some ncORFs are subject to evolutionary constraints. This is not surprising. The authors should provide further analyses on more detailed features of these "conserved" ncORFs vs. the "non-conserved" ones. Some pretty informative works have been done in *Drosophila*, worms, mice, and humans. Figure 3 suggests some ncORFs are under evolutionary constraint, but this is not unexpected. More granular analyses contrasting "conserved" versus "non-conserved" ncORFs would be informative. In fact, small ORFs, especially uORFs, have been extensively studied for their functions and cross-species conservation. The authors should explicitly show what is new here in their analyses.
- (5) Translation levels are reported using RPF counts. However, translation efficiency (normalized by RNA expression) is a more appropriate measure to account for expression heterogeneity.
- (6) The correlation analyses between ncORF translation levels and PhyloCSF are confusing and largely descriptive. These sections need sharper framing and clearer conclusions.
- (7) Public ribo-seq datasets, generated by different research labs, are known for their strong batch effects. Representations of tissues and cells are also very unbalanced. Therefore, the co-translation analysis between ncORFs and canonical CDS is not well controlled. This should be done by referring to a recent large-scale ribo-seq meta-analysis (Nat Biotechnol. 2025. doi: 10.1038/s41587-025-02718-5).

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Reviewer #2 (Public review):

Summary:

Chang et al. attempted to analyze a large number of ribo-seq datasets through a standardized pipeline, identifying novel non-canonical ORFs and elucidating their evolutionary and expression characteristics.

Strengths:

- (1) The datasets analyzed by the authors are sufficiently comprehensive, and the use of standardized pipelines ensures excellent analytical consistency.
- (2) Their analyses of ORF evolution and co-expression further deepen our understanding of these ORFs.

Weaknesses:

(1) The authors primarily conducted analyses through bioinformatics, lacking sufficient wet-lab experimental evidence.

(2) Regarding the evolution of non-canonical ORFs, a considerable amount of prior work already exists. The authors need to further clarify what new insights and discoveries they have made based on the analysis of such a large dataset.

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